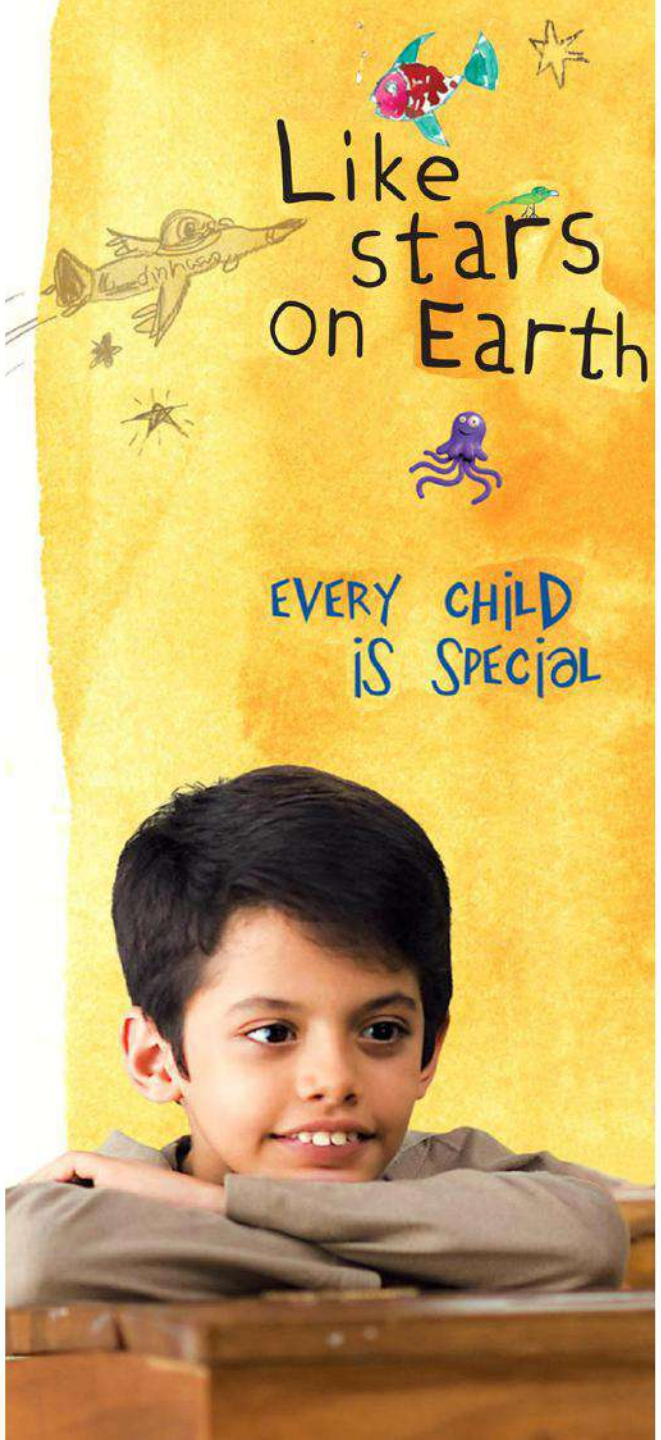


SCREENING AND REFINEMENT SYSTEM TO IDENTIFY THE RISK OF DYSCALCULIA AND DYSGRAPHIA LEARNING DISABILITIES IN PRIMARY SCHOOL STUDENTS

20_21-J 01

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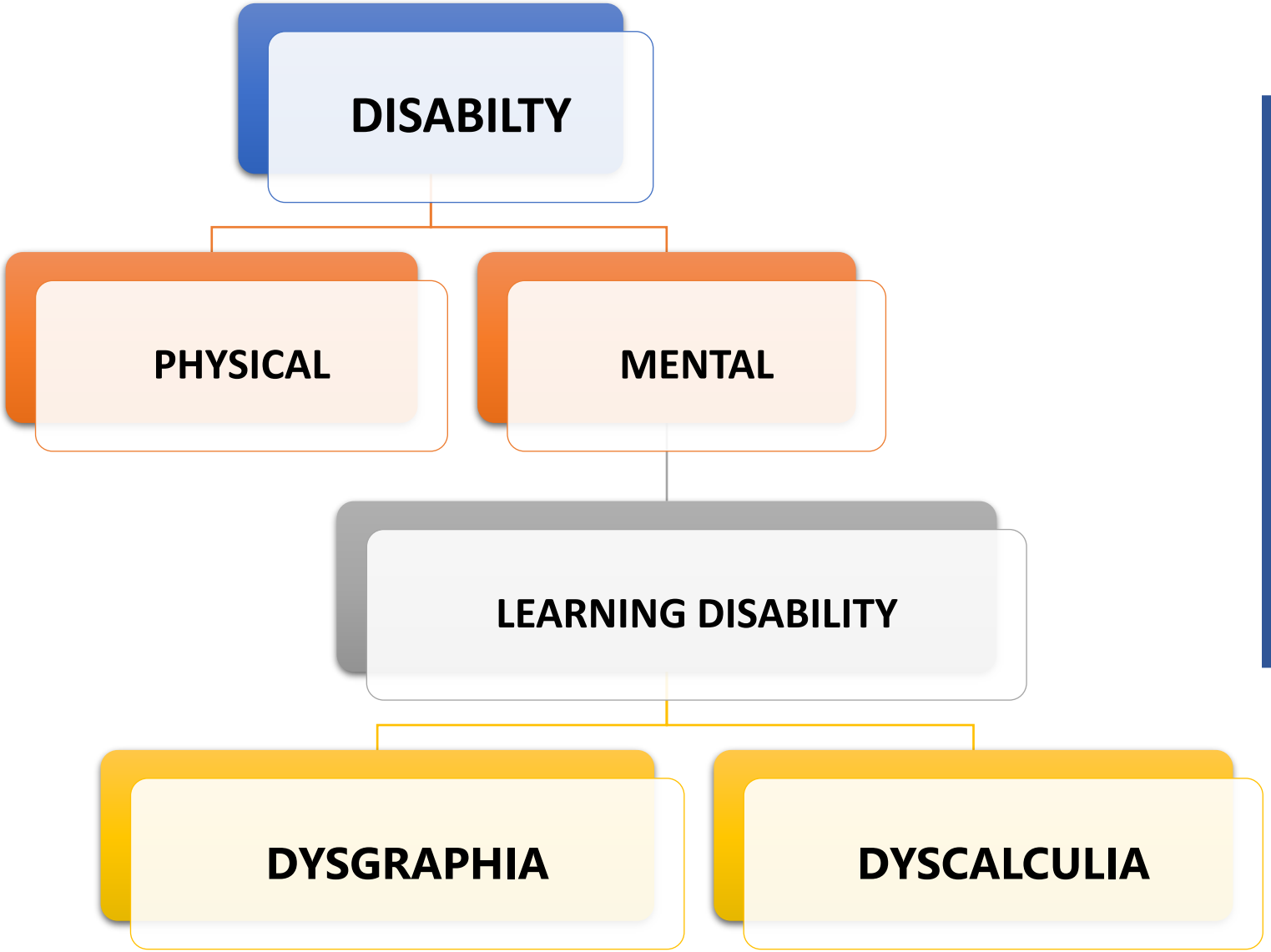


Liyanage T.J.P.S.

Content

1. Introduction
2. Research Question
3. Objectives
4. System Overview Diagram
5. Technologies
6. Individual Progress
7. Standards and Best Practices
8. Non-Functional Requirements
9. Commercialization Aspects
10. Budget

1. INTRODUCTION





INTRODUCTION CONT.



NANA SHILPA

DYSGRAPHIA

Difficulty of writing in provided space



Unclear and irregular handwriting



Slow writing pace



DYSCALCULIA



Mathematics disorder



Difficulty in identifying numbers



Difficulty in perform basic mathematical operations

2. RESEARCH QUESTION

How to provide mobile application-based solution to

- Screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities(LD) in primary school students ?
- Deliver Comprehensive activities to overcome those LDs risks?





3. OBJECTIVES

MAIN OBJECTIVE

- Provide mobile application-based solution to screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students.

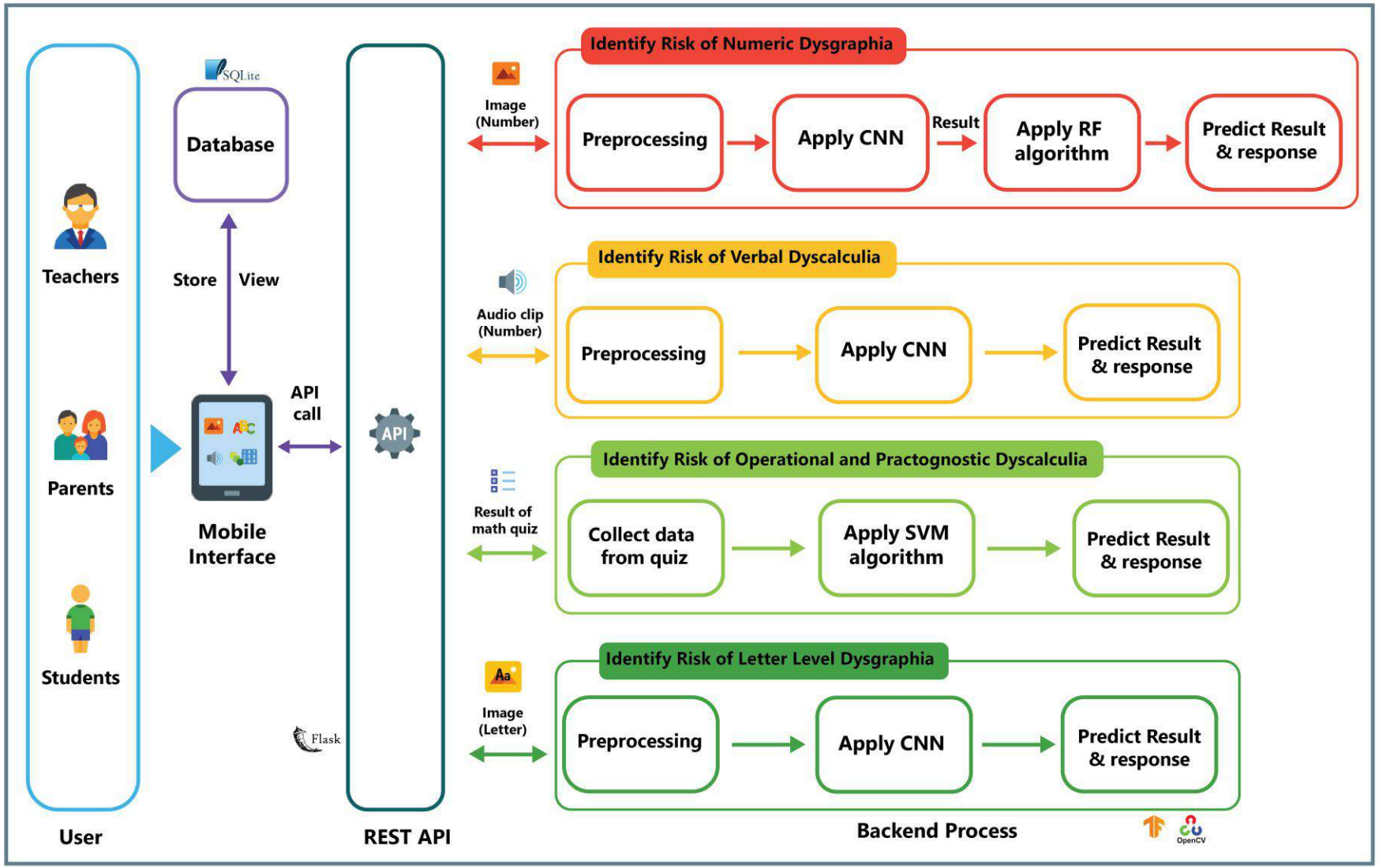


SPECIFIC OBJECTIVES

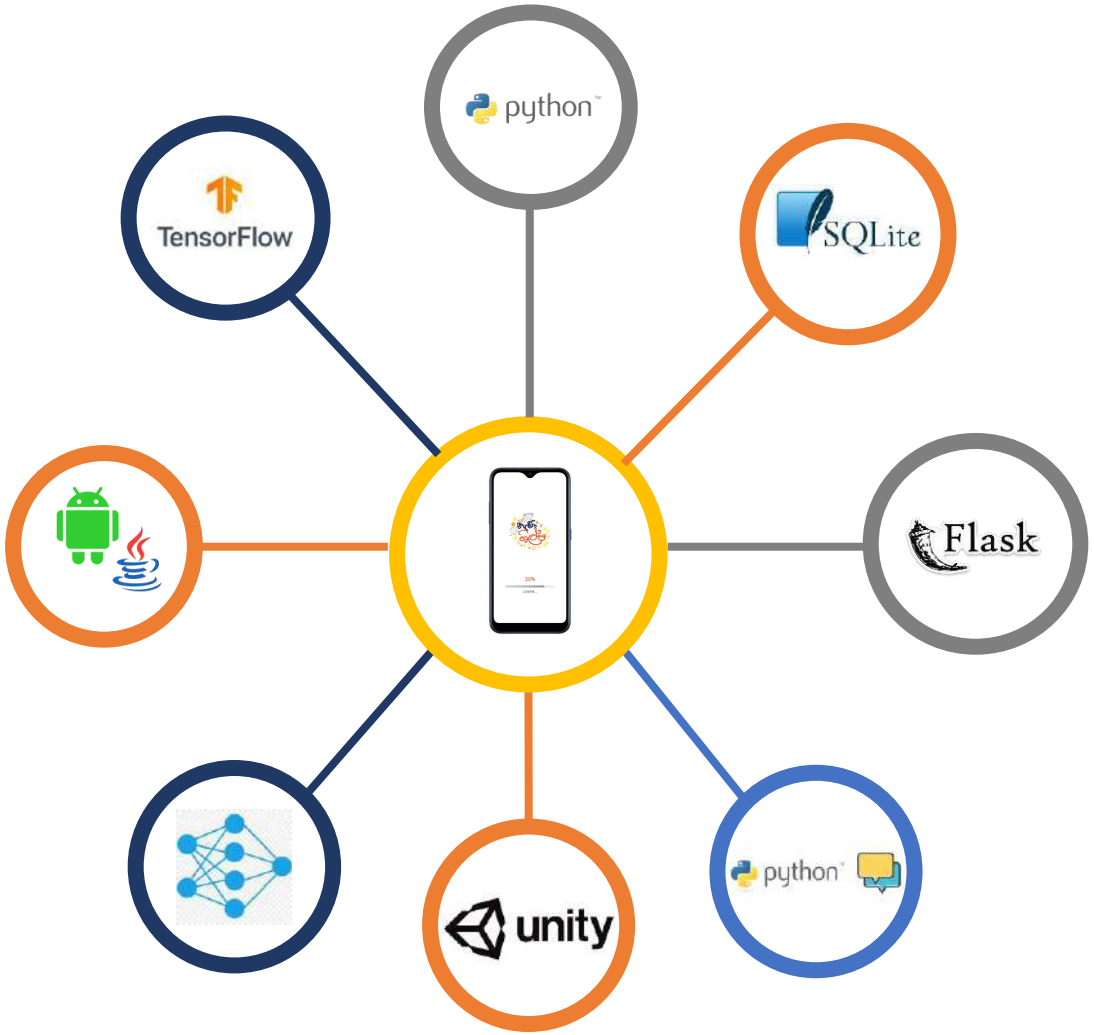


- Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.
- Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.
- Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition.

4. SYSTEM OVERVIEW DIAGRAM



5. TECHNOLOGIES



Mobile Application

- Android

Database

- SQLite

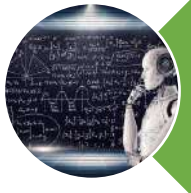
Middle Ware Technologies

- Python
- TensorFlow
- REST API - Flask



Technical Concepts

- Machine Learning
- Convolutional Neural Network (CNN)
- Image Processing



Algorithms

- Support Vector Machine
- FFT Algorithm



IT17180108 | Kalpani Sandupaba Abeysinghe A.D.

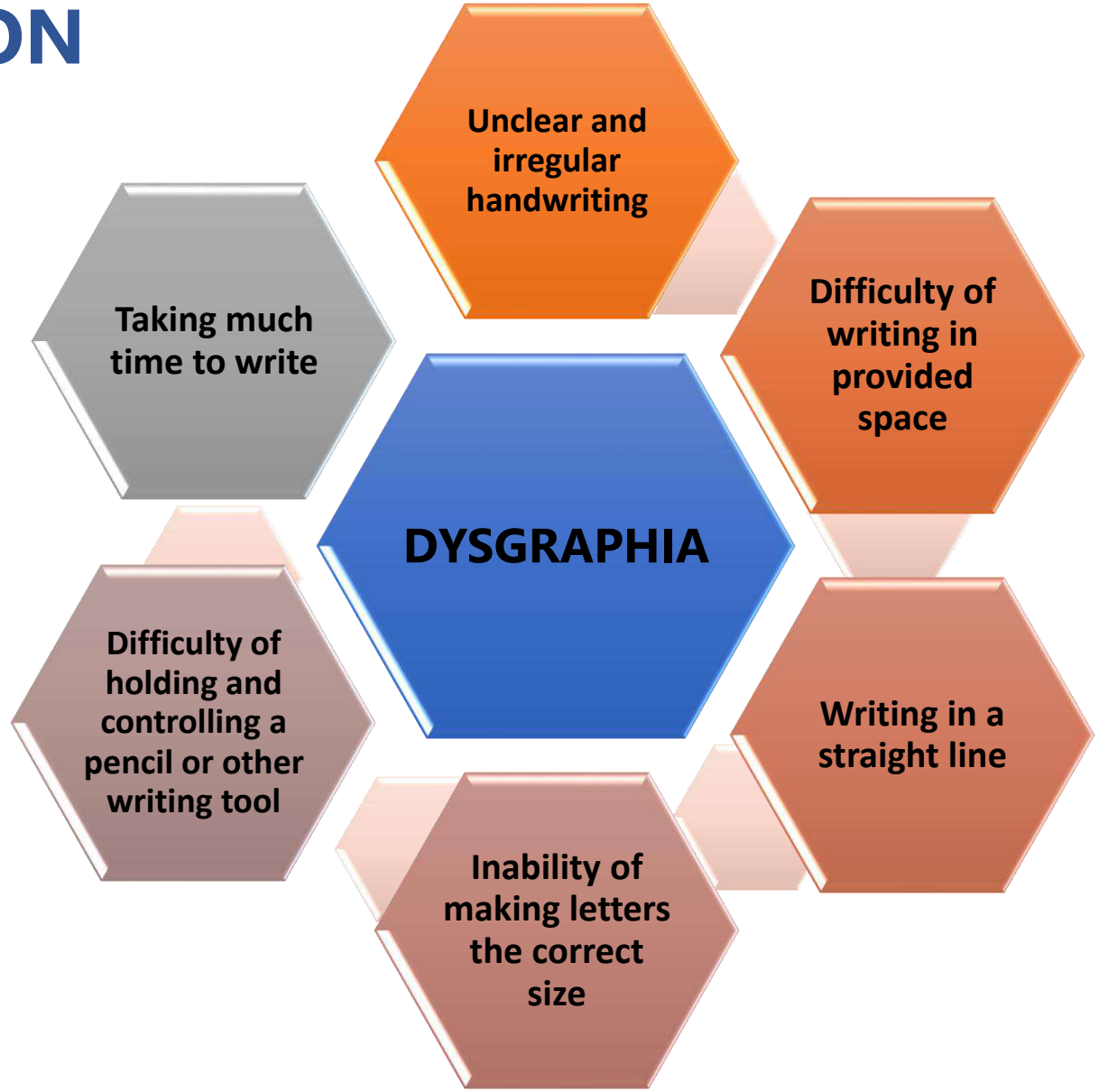
BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

THE PROGRESS

Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition

❖ INTRODUCTION

❖ BACKGROUND



❖ RESEARCH GAP

Assistive Learning Environment for Students with Learning Disabilities[3]

- Proposed Assistive Learning Environment (ALE) for the students who are undergoing with learning disabilities.
- Provide environment for the students in order to follow the activities in their own pace.

A Mobile Web Game Approach for Improving Dysgraphia[4]

- Provide game-based solution for dysgraphia students.
- Implemented using client server web application architecture.

An Expert System for Diagnosing Dysgraphia [5]

- Identify the Dysgraphia condition through the decision tree analyze techniques.

Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

- System for the teachers and students where teacher can view all the class through on interface.
- Based on the initial conducted activities, improvement learning model is assigned by the teacher.

❖ SYSTEM COMPARISON

Features	01	02	03	04	Nena Shilpa
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Letter Identification	✗	✗	✗	✗	✓
Supports native Sinhala language	✗	✗	✗	✗	✓
Recover activities based on the risk level	✗	✗	✗	✓	✓
Keep track of the progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
User attractive interfaces	✗	✓	✗	✓	✓

01-Assistive Learning Environment for Students with Learning Disabilities[3]
02 -A Mobile Web Game Approach for Improving Dysgraphia[4]
03 -An Expert System for Diagnosing Dysgraphia [5]
04 -Automatic segmentation as a tool for examining the handwriting process of children with Dysgraphic and proficient handwriting [6]

❖ RESEARCH PROBLEM

How to provide mobile application-based solution to

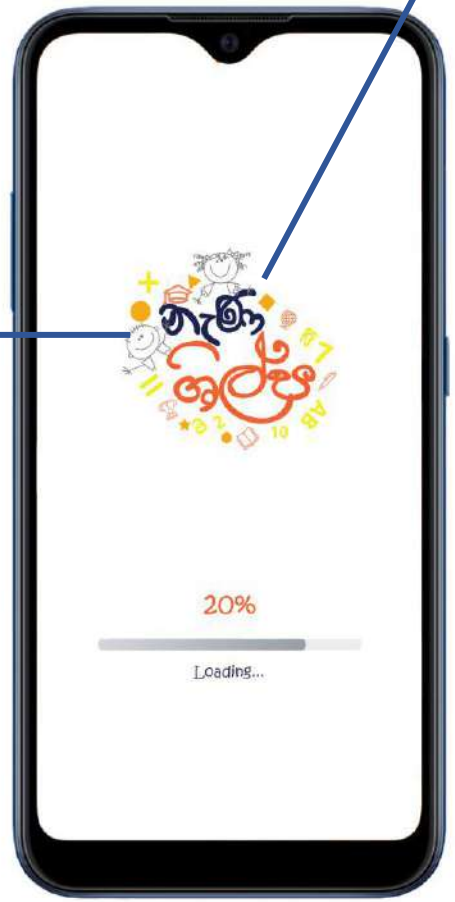
- **Identify the risk of letter level Dysgraphia condition ?**
- **Deliver risk diminish activities ?**
- **Monitor the betterment of the students ?**



❖ OBJECTIVES

SPECIFIC OBJECTIVE

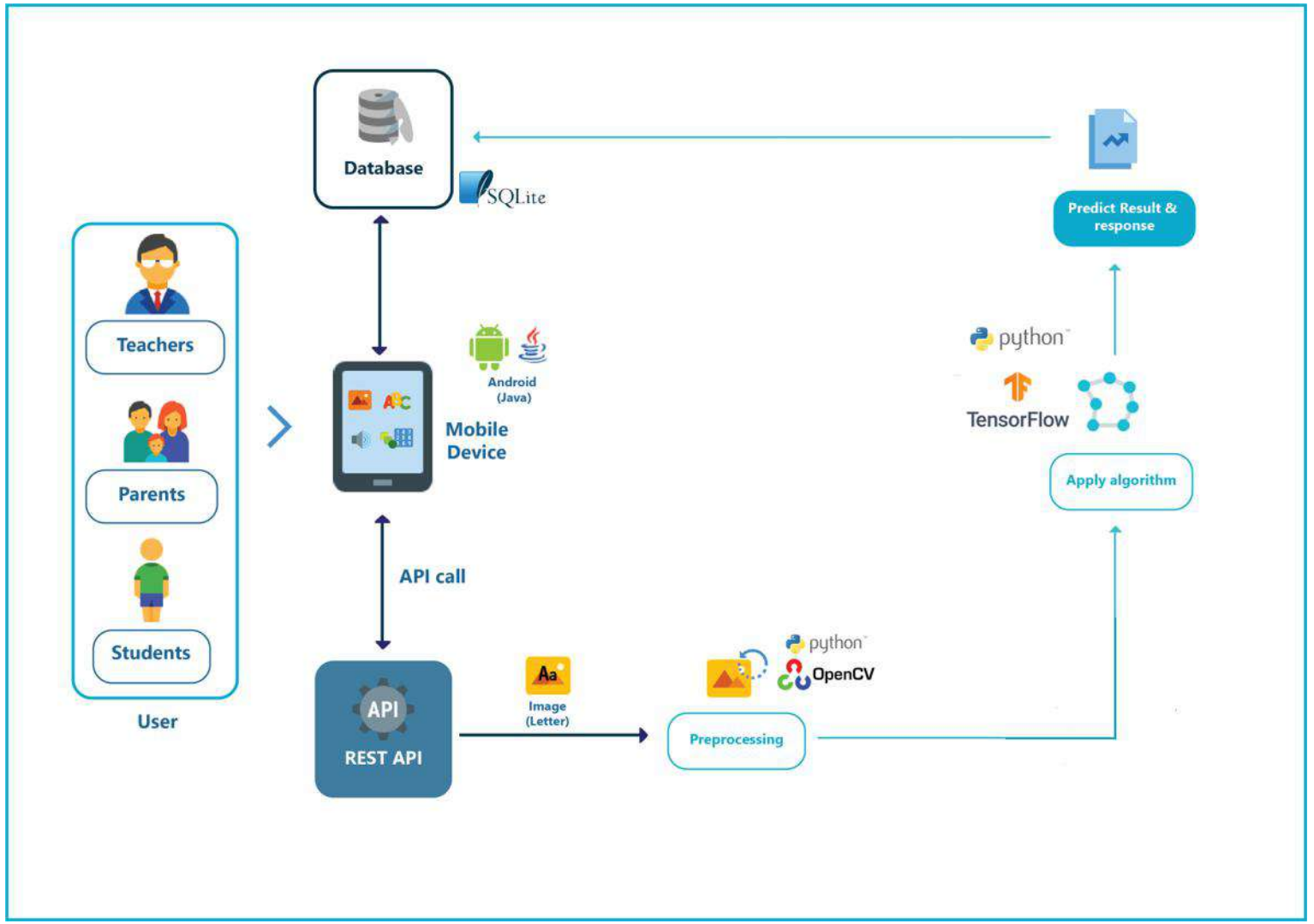
❖ Identify the risk of Letter level Dysgraphia through Sinhala Characters and provide activities to improve the condition.



Sub Objectives

- Letter level Dysgraphia risk identification.
- Practice activities to improve the condition.
- Keep the track of student activities.
- Report generation for the performed test activities.

METHODOLOGY



Component Overview Diagram

❖ KEY PILLARS


Deep Learning
Trained Data Model
Machine Learning
Conventional Neural Network
Image Processing

❖ ACHIEVEMENTS



- Successful Data Collection for Model Training
- Completed Primary UI Design and Implementation
- User Input Collection Mechanism Implementation
- ML model has been trained with Accuracy 98%
- Backend Endpoint Implementation and Response Handling

❖ ACHIEVEMENT EVIDENCE



Sub Menu



Screening UI for Letter and Words



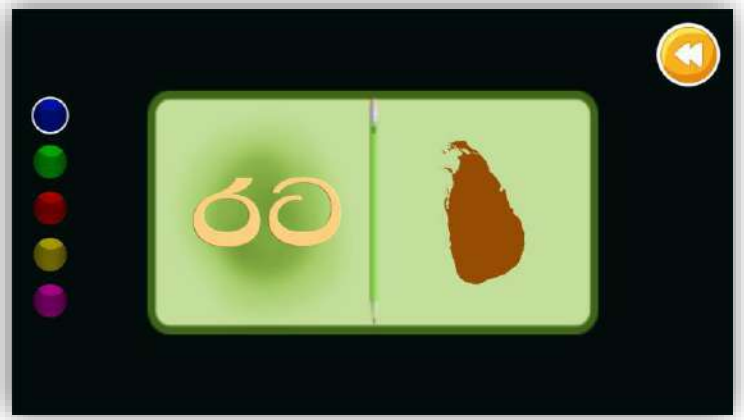
Sub Menu for Intervention



Sub Menu for Word Intervention



Letter Intervention Practice UI

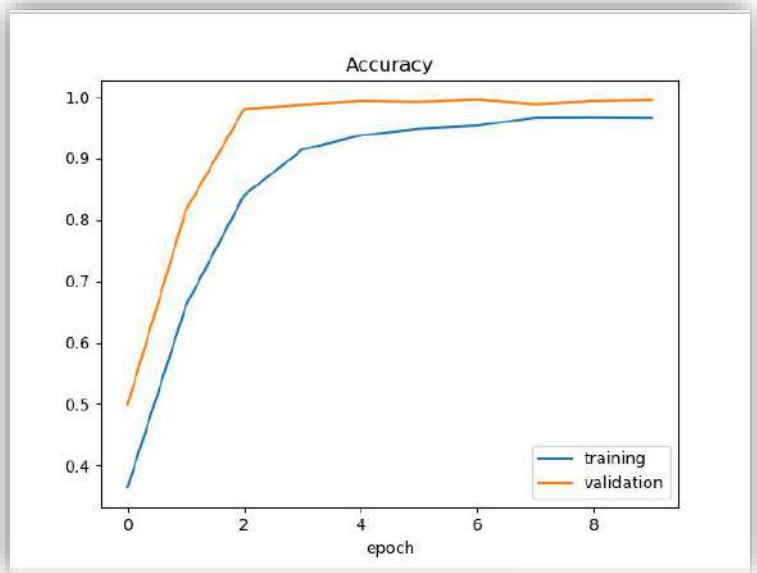


Word Intervention Result UI

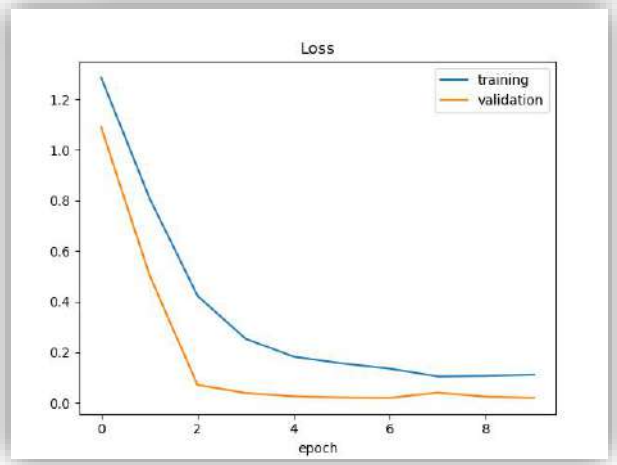
❖ RESULTS AND DISCUSSION

```
Epoch 1/10
109/109 [==============================] - 16s 144ms/step - loss: 1.2859 - accuracy: 0.3645 - val_loss: 1.0893 - val_accuracy: 0.4993
Epoch 2/10
109/109 [==============================] - 16s 145ms/step - loss: 0.8111 - accuracy: 0.6597 - val_loss: 0.5072 - val_accuracy: 0.8151
Epoch 3/10
109/109 [==============================] - 14s 131ms/step - loss: 0.4237 - accuracy: 0.8398 - val_loss: 0.0716 - val_accuracy: 0.9803
Epoch 4/10
109/109 [==============================] - 15s 136ms/step - loss: 0.2542 - accuracy: 0.9140 - val_loss: 0.0392 - val_accuracy: 0.9868
Epoch 5/10
109/109 [==============================] - 15s 137ms/step - loss: 0.1824 - accuracy: 0.9371 - val_loss: 0.0260 - val_accuracy: 0.9927
Epoch 6/10
109/109 [==============================] - 15s 141ms/step - loss: 0.1563 - accuracy: 0.9478 - val_loss: 0.0213 - val_accuracy: 0.9912
Epoch 7/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1354 - accuracy: 0.9528 - val_loss: 0.0200 - val_accuracy: 0.9949
Epoch 8/10
109/109 [==============================] - 16s 144ms/step - loss: 0.1047 - accuracy: 0.9660 - val_loss: 0.0407 - val_accuracy: 0.9876
Epoch 9/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1067 - accuracy: 0.9664 - val_loss: 0.0247 - val_accuracy: 0.9927
Epoch 10/10
109/109 [==============================] - 16s 147ms/step - loss: 0.1115 - accuracy: 0.9657 - val_loss: 0.0200 - val_accuracy: 0.9942
Test Score =  0.02710871584713459
Test Accuracy = 0.9929783344268799

Process finished with exit code 0
```



CNN Model Accuracy



❖ TEST CASES

Test case ID	NS001
Test Case Scenario	Take the screening test to identify whether the user has the risk of having letter level dysgraphia
Test input data	User Written letters
Test procedure	1. Selects Sinhala Writing (සිංහල ලියන්න) button in main menu 2. Select Sinhala Writing (සිංහල ලියන්න) button in sub menu to take the test. 3. Write the letter or word which is displayed in the banner. (අ, ල, ර, රට, මල) 4. After writing the letter, click go forward button (ඉදිරියට යමු) to submit the answer. 5. After successful completion of the writing letters and words, user will be redirected to the result page.
Expected output	User should be getting the messages of accuracy percentage and result of having risk of letter level dysgraphia.
Actual output	Get the message of not having the risk of letter level dysgraphia.
Test result	Pass.

❖ TEST CASES CONT.

Test case ID	NS002
Test Case Scenario	Take the intervention practice test to in order to overcome risk of having letter level dysgraphia
Test input data	User Written letter or Word in dashed interface
Test procedure	1. Selects Sinhala Writing (සිංහල ලියන්න) button in main menu 2. Select Sinhala class (සිංහල පන්තිය) button in sub menu to take the practice test. 3. Select Sinhala Letter Practice (අකුරු හුරුව) button to take letter practice test. 4. Select the preferred letter from four letters available in the menu. 5. Write the letter inside the dashed lined with following the indexes. 6. After successful completion of the writing letter (filling the letter correctly inside of the dashed lines), user will encounter with the success pop up with letter and picture.
Expected output	User encounter with the success pop up with letter and picture.
Actual output	User has encounter with the success pop up with letter and picture.
Test result	Pass.

❖ REFEREANCES

[01]Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page

[02]Mishna, F. (2003). Learning Disabilities and Bullying. *Journal of Learning Disabilities, 36*(4), 336-347. doi:10.1177/00222194030360040501

[03]Z. Pirani and M. Sasikumar, "Assistive learning environment for students with Learning Disabilities," *2012 4th International Conference on Intelligent Human Computer Interaction (IHCI)*, 2012

.

[04] D. Giordano and F. Maiorana, "A Mobile Web Game Approach for Improving Dysgraphia," Proceedings of the 7th International Conference on Computer Supported Education, 2015.

[05] D. A. Kurniawan, S. W. Sihwi, and Gunarhadi, "An expert system for diagnosing dysgraphia," 2017 2nd International conferences on Information Technology, Information Systems and Electrical Engineering (ICITISEE), 2017.

[06] Rosenblum, Sara & Dvorkin, Assaf & Weiss, Patrice. (2006). Automatic segmentation as a tool for examining the handwriting process of children with dysgraphic and proficient handwriting. Human movement science. 25. 608-21. 10.1016/j.humov.2006.07.005.



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THE PROGRESS

Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.

INTRODUCTION

❖ BACKGROUND

- A special Learning Disability called **Numeric Dysgraphia (ND)** not a very common term.
- This disability causes difficulties when writing the numbers.
- No special tests or treatments are not identified and cannot be cured [1].
- This kind of students can be identified by their numeric writing patterns by the teacher or a parent who pays much attention.

Through activities / practices, the situation of the student can be improved to some extent. So, it is important to identify this kind of students at early stage.



RESEARCH GAP

IO-CRIVO mobile app[1]

- For the students who has difficulties in writing.
- Screening for Dysgraphia
- PHP, HTML, CSS, JavaScript, MySQL

An Expert System[2]

- Provide remedy for Dysgraphic students on their studies
- A writing task and questionnaire was provided.

AR-DAWE model[3]

- AR based Dysgraphia Assistance Writing Environment (AR-DAWE).
- Identify spelling mistakes of the students.

A writing questionnaire[4]

- A statistical model
- A questionnaire

SYSTEM COMPARISON

Features	[1]	[2]	[3]	[4]	Nena Shilpa
Screening or intervention for Dysgraphia	✓	✓	✓	✓	✓
Screening and intervention for Dysgraphia	✗	✗	✗	✗	✓
Screening for Numeric Dysgraphia	✗	✗	✗	✗	✓
Remedies for ND in activity-based environment	✗	✗	✗	✗	✓
Keep track of the progress	✓	✗	✗	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases	✗	✗	✓	✗	✓

RESEARCH PROBLEM

How to develop a solution which is a mobile application,



To identify the risk for ND ?



To provide remedies ?



To keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

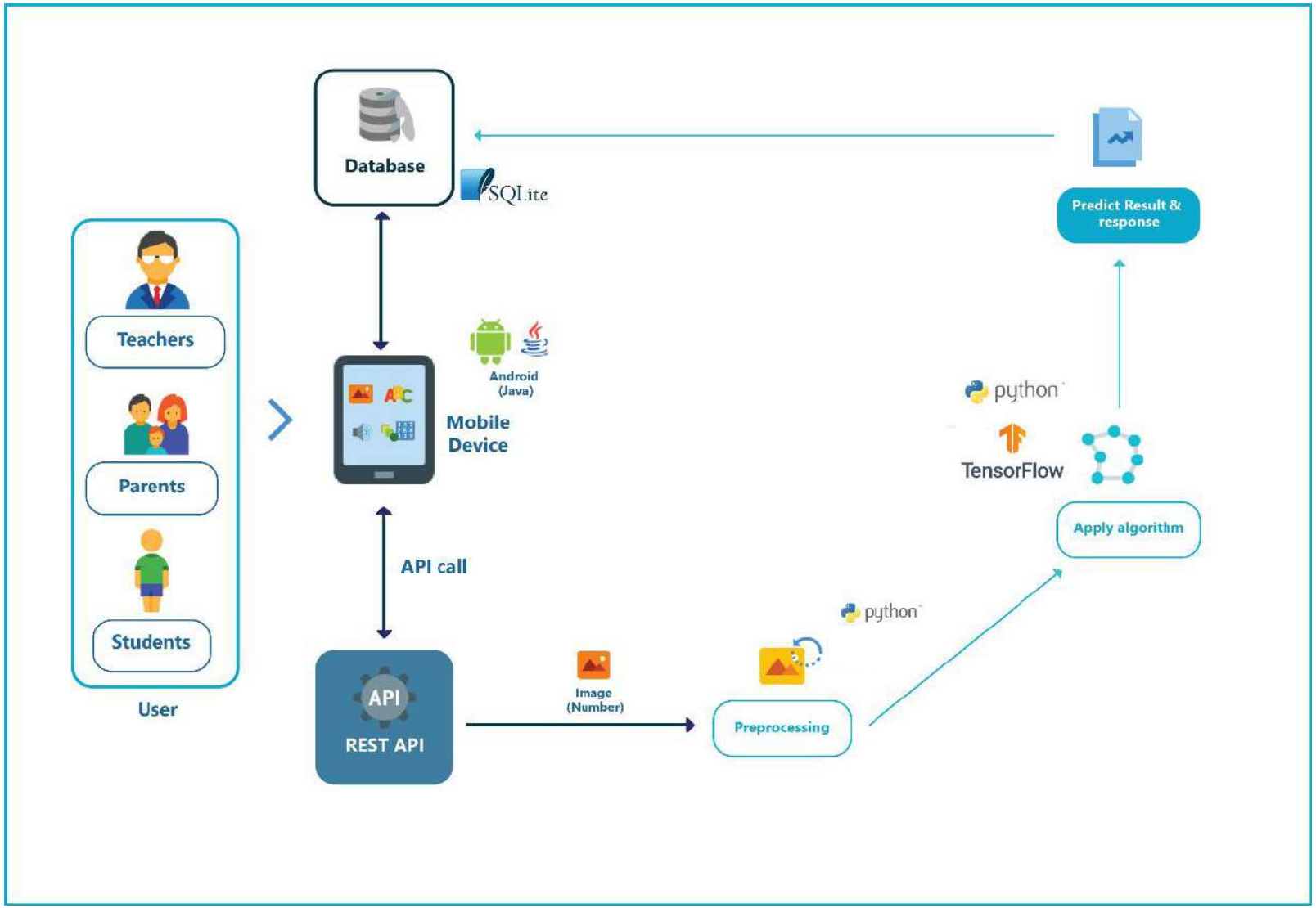
- Identify the risk for Numeric Dysgraphia and provide remedies while keeping the track of progress.



Sub Objectives

- Screening for the risk for Numeric Dysgraphia
- Providing activity-based treatments
- Keep the track of progress

METHODOLOGY



System Component Overview Diagram

KEY PILLARS



Machine Learning



Tensor Flow Lite

Random Forest Algorithm

Deep Learning



Convolutional Neural Network





❖ ACHIEVEMENTS



- Data gathered successfully.
- Kept the track of progress.
- Successfully trained the CNN 99% accuracy percentage.
- Achieved 99% accuracy percentage in ML model training.
- UI/UX designs

❖ ACHIEVEMENTS EVIDENCES



Screening UI



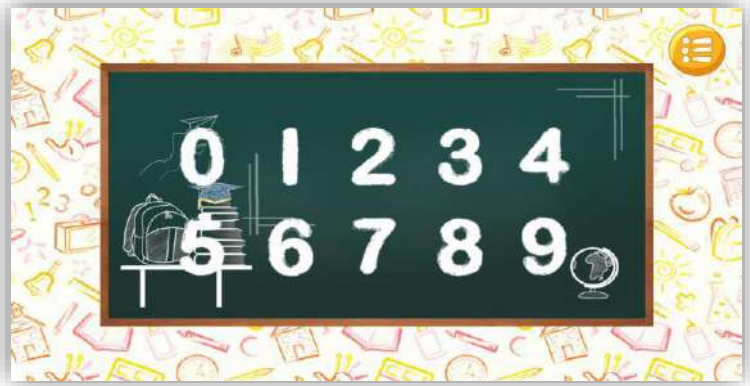
Current Quiz Result UI



Show Results Stages UI



Show Results Stages with details UI



Intervention selection UI

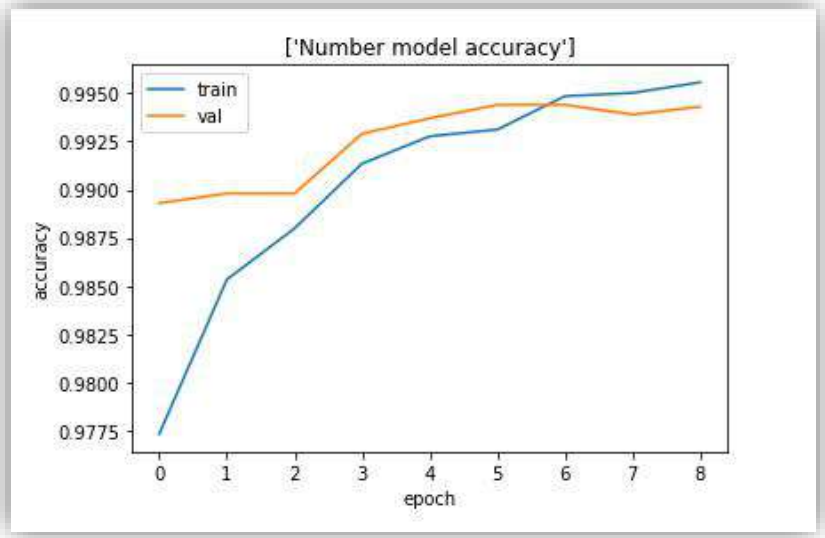


Intervention UI

❖ RESULTS AND DISCUSSION

```
Model: "model_1"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
input (InputLayer)           [(None, 28, 28, 1)]       0
_________________________________________________________________
conv2d_3 (Conv2D)            (None, 23, 23, 24)        888
_________________________________________________________________
batch_normalization_4 (Batch (None, 23, 23, 24)        72
_________________________________________________________________
activation_4 (Activation)    (None, 23, 23, 24)        0
_________________________________________________________________
dropout_4 (Dropout)          (None, 23, 23, 24)        0
_________________________________________________________________
conv2d_4 (Conv2D)            (None, 10, 10, 48)        28848
_________________________________________________________________
batch_normalization_5 (Batch (None, 10, 10, 48)        144
_________________________________________________________________
activation_5 (Activation)    (None, 10, 10, 48)        0
_________________________________________________________________
dropout_5 (Dropout)          (None, 10, 10, 48)        0
_________________________________________________________________
conv2d_5 (Conv2D)            (None, 4, 4, 64)          49216
_________________________________________________________________
batch_normalization_6 (Batch (None, 4, 4, 64)          192
_________________________________________________________________
activation_6 (Activation)    (None, 4, 4, 64)          0
_________________________________________________________________
dropout_6 (Dropout)          (None, 4, 4, 64)          0
_________________________________________________________________
flatten_1 (Flatten)          (None, 1024)              0
_________________________________________________________________
dense_1 (Dense)              (None, 200)               205000
_________________________________________________________________
batch_normalization_7 (Batch (None, 200)               600
_________________________________________________________________
activation_7 (Activation)    (None, 200)               0
_________________________________________________________________
dropout_7 (Dropout)          (None, 200)               0
_________________________________________________________________
output (Dense)               (None, 10)                2010
=================================================================
Total params: 286,970
Trainable params: 286,298
Non-trainable params: 672
_________________________________________________________________
```

```
313/313 [==============================] - 4s 12ms/step - loss: 0.0171 - accuracy: 0.9943
Test error:  0.01714743860065937, 'Test Accuracy: ' 0.9943000078201294
```



Trained CNN model Accuracy

❖ RESULTS AND DISCUSSION Cont..

```
]: experiment_accuracy = sm.accuracy_score(y_test, prediction_y)
   print('Accuracy score is : ', str(experiment_accuracy))

   Accuracy score is :  0.9896907216494846
```

Accuracy of Trained ML Model

```
Classification report:                    precision    recall  f1-score   support

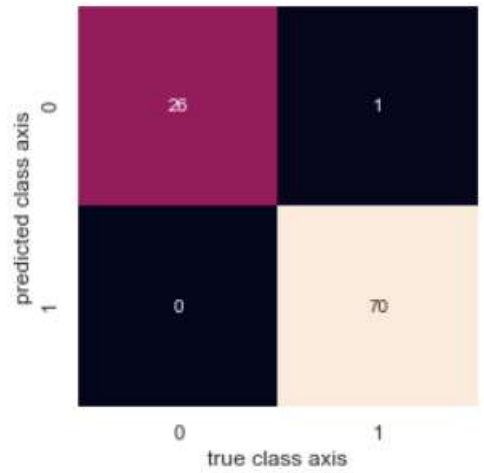
   Numeric Dysgraphia       1.00      0.96      0.98        27
No Numeric Dysgraphia       0.99      1.00      0.99        70

             accuracy                           0.99        97
            macro avg       0.99      0.98      0.99        97
         weighted avg       0.99      0.99      0.99        97
```

Classification Report of Trained ML Model

```
: sb.heatmap(confusionmt.T, square = True, annot = True, fmt = 'n', cbar = False)
  pt.xlabel('true class axis')
  pt.ylabel('predicted class axis')

: Text(89.18, 0.5, 'predicted class axis')
```



Show Test Results Using a Diagram

❖ TEST CASES

Test case ID	NS003
Test Case Scenario	Screening for Numeric Dysgraphia – for normal primary school student.
Test input data	Bitmap version of written numbers by the student.
Test procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in main menu. 2. Selects the button for math class (ගණිත පන්තිය) in sub main menu. 3. Starts to write numbers from 0 to 9 each twice. 4. App displays the results of the done quiz. 5. Clicks the button (හඳුනාගන්න) to predict the Numeric dysgraphia situation. 6. App displays the results based on quiz results.
Expected output	App should display as non-dysgraphic situation.
Actual output	App displays as non-dysgraphic situation.
Test result	Pass.



❖ TEST CASES CONT …

Test case ID	NS003
Test Case Scenario	Screening for Numeric Dysgraphia – for diagnosed primary school student.
Test input data	Bitmap version of written numbers by the student.
Test procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in main menu. 2. Selects the button for math class (ගණිත පන්තිය) in sub main menu. 3. Starts to write numbers from 0 to 9 each twice. 4. App displays the results of the done quiz. 5. Clicks the button (හඳුනාගන්න) to predict the Numeric dysgraphia situation. 6. App displays the results based on quiz results.
Expected output	App should display as dysgraphic situation.
Actual output	App displays as dysgraphic situation.
Test result	Pass.



❖ REFERENCES

[1]. Giordano, D. & Maiorana, Francesco. (2014). Addressing dysgraphia with a mobile, web-based software with interactive feedback. 2014 IEEE-EMBS International Conference on Biomedical and Health Informatics, BHI 2014. 264-268. 10.1109/BHI.2014.6864354.

[2]. Kurniawan, Dimas & widya sihwi, Sari & Gunarhadi, Gunarhadi. (2017). An expert system for diagnosing dysgraphia. 468-472. 10.1109/ICITISEE.2017.8285552.

[3]. Khan, Muhammad & Hussain, Muhammad & Ahsan, Kamran & Saeed, Muhammad & Nadeem, Adnan & Ali, Syed & Mahmood, Nadeem & Rizwan, Kashif. (2017). Augmented Reality Based Spelling Assistance to Dysgraphia Students. Journal of Basic & Applied Sciences. 13. 500-507. 10.6000/1927-5129.2017.13.82.

[4]. Rosenblum, Sara & Dror, Gideon. (2016). Identifying Developmental Dysgraphia Characteristics Utilizing Handwriting Classification Methods. IEEE Transactions on Human-Machine Systems. PP. 1-7. 10.1109/THMS.2016.2628799.

[5]. 5 Best Machine Learning Algorithms for Classification Problems [online] Available: http://www.intellspot.com/machine-learning-classification/



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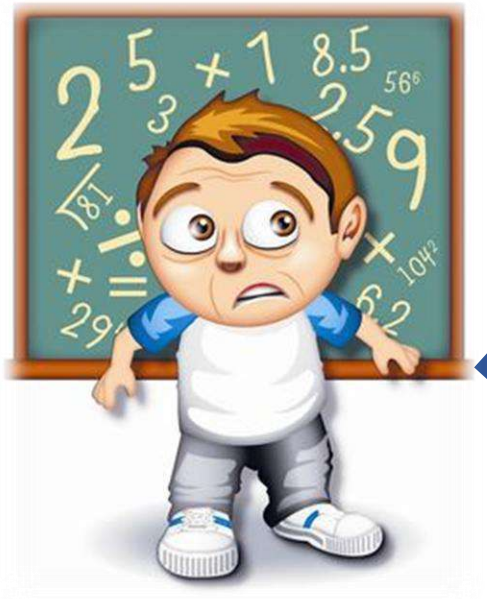
THE PROGRESS

Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.

INTRODUCTION

❖ Background

- ❖ Dyscalculia or mathematics learning disability it is a specific learning disability directed to arithmetic.[1][2].
- ❖ Dyscalculia is sometimes called as number blindness.
- ❖ The prevalence of developmental dyscalculia is between 3% and 6% in the school population.
- ❖ Among the dyscalculia types two of them are focusing on here,
  - **Operational Dyscalculia**
  - **Practognostic Dyscalculia**



RESEARCH GAP

Screening for Disorders of Mathematics via a web application [3]

- Learning process with calculation, numeric concepts, and problem solving.
- HTML5, CSS, JavaScript while using the MySQL as the database.

Development of Assistive Technology for Students with Dyscalculia [4]

- Provided Solution as a software tool which named as "Calculating Aid Tools: KidKanit" which helps to dyscalculia students in mathematical area.

Web-based learning tool for primary school student with Dyscalculia [5]

- Cover the most common characteristics which were the Dyscalculia students struggle with in specific areas such as counting, number recognition.

Screening for Dyscalculia within HE (Higher Education) [6]

- Provide screening process to identify dyscalculia in higher education students.
- In their solution, it was made a mobile app called DyscalculiUM.

SYSTEM COMPARISON

Features	[3]	[4]	[5]	[6]	Nena Shilpa
Dyscalculic Identification.	✗	✗	✗	✓	✓
Provide activities to improve the math concepts.	✓	✓	✓	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✓
User attractive interfaces with sound bases (multi-scenery)	✗	✓	✗	✗	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✗	✗	✗	✓
Supervised Learning(uses Machine Learning)	✗	✗	✗	✗	✓

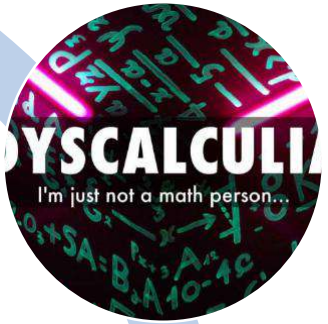




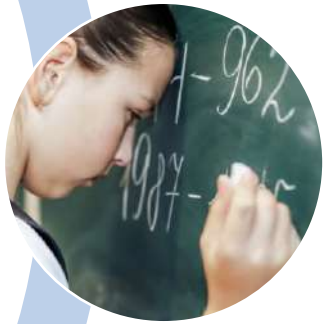
- Screening for Disorders of Mathematics via a web application [3]
- Development of Assistive Technology for Students with Dyscalculia [4]
- Web-based learning tool for primary school student with Dyscalculia [5]
- Screening for Dyscalculia within HE (Higher Education) [6]

RESEARCH PROBLEM

How to develop mobile **application-based solution to**



- Identify the risk for Operational and Practognostic Dyscalculia ?



- Provide activities to improve the condition ?



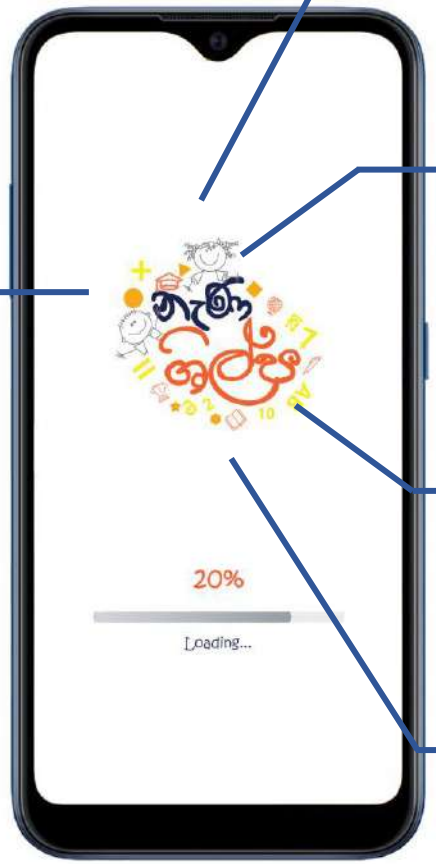
- Keep the track of progress ?



OBJECTIVES

SPECIFIC OBJECTIVE

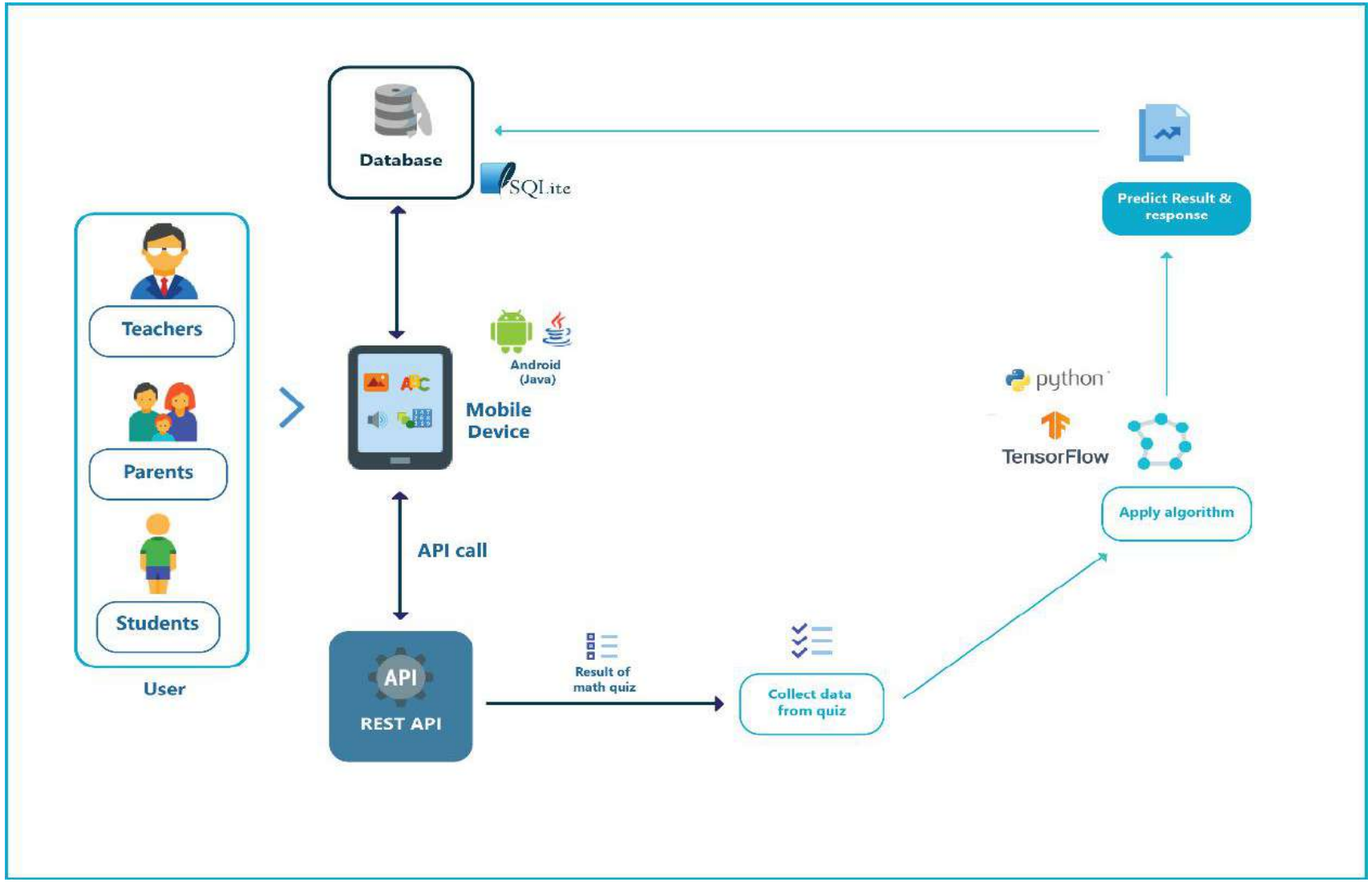
- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.



Sub Objectives

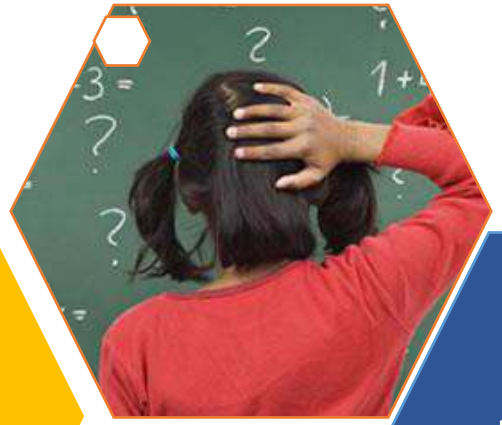
- Identify risk of Operational and Practognostic Dyscalculia.
- Provide activities for practice to improve the condition.
- Keep the track of student progress.
- Create Screening model

RESEARCH METHODOLOGY



Component Overview Diagram

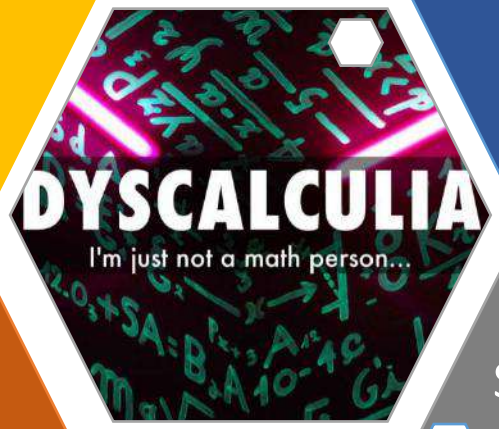
KEY PILLARS



Machine Learning

Data mining







Supervised Learning

Support Vector Machine with RBF kernel

ACHIEVEMENTS



- Data Collection for Model Training
- Screening part of the Identify risk of Operational and Practognostic Dyscalculia (Created an interactive math quiz)
- Activities for practice to improve the condition (remedies).
- Keep track Student progress.
- ML model has been trained with 98% Accuracy.
- Implemented Backend Endpoints with Python Flask server to make RESTful API call.

ACHIEVEMENT EVIDENCE



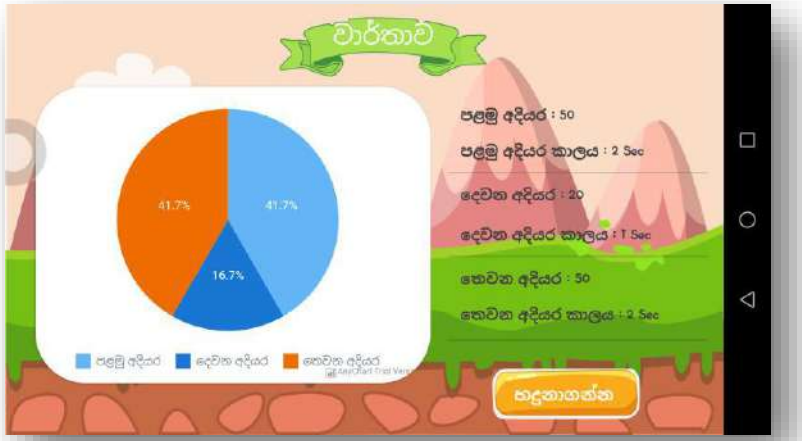
Stage 1



Stage 2



Stage 3



Result



Report



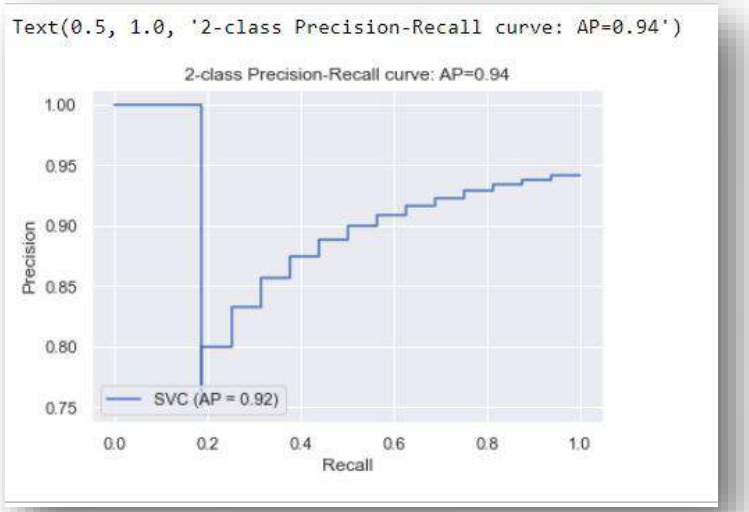
Remedy Activities

RESULTS & DISCUSSION

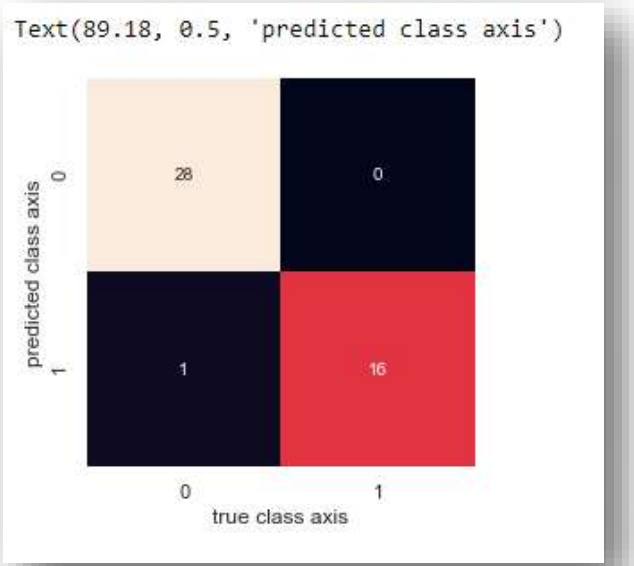
Classification Report :

	precision	recall	f1-score	support
No Dyscalculia	0.97	1.00	0.98	28
Dyscalculia	1.00	0.94	0.97	17
accuracy			0.98	45
macro avg	0.98	0.97	0.98	45
weighted avg	0.98	0.98	0.98	45

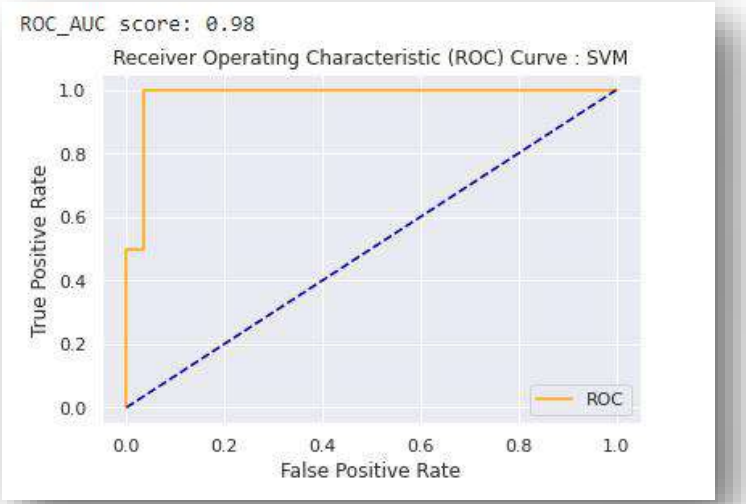
Classification report



Precision recall Curve for Dyscalculia Screening



Test Prediction Result



ROC curve for dyscalculia identification

❖ TEST CASES

Test case ID	TC01
Test Case Scenario	Predict Operational and Practognostic dyscalculia – User done mathematical quiz accurately (for normal primary school student).
Test input data	Mathematical quiz results (90, 4.32, 100, 2.54, 90, 7.35).
Test procedure	1. Select Math problem (ගණිත ගැටලු) button in main menu. 2. Select Math class (ගණිත පන්තිය) button in sub menu in order to do the math quiz. 3. Proceed with the enumerating quiz as stage 1 and click next button in dialog box after completion. 4. Proceed with the comparison objects and numbers quiz as stage 2 and click next button in dialog box after completion. 5. Proceed with the mathematical operation-based quiz covering the addition and subtraction quiz as stage 3 and click next button in dialog box after completion. 6. Display the results of each stages and average time to answer each question in result. 7. Click predict (හඳුනාගන්න) button. 8. Display whether children who suffer from dyscalculia or non- dyscalculia (සමත් or අසමත්).
Expected output	User Should be notify as non- dyscalculic child (සමත්)
Actual output	User is notified as non- dyscalculia child (සමත්)
Test result	Pass



❖ TEST CASES Cont.

Test case ID	TC02
Test Case Scenario	Predict Operational and Practognostic dyscalculia – User done mathematical quiz weakly (for dyscalculic child)
Test input data	Mathematical quiz results (40, 11.42, 50, 8.56, 30, 14.59).
Test procedure	1. Select Math problem (ගණිත ගැටලු) button in main menu. 2. Select Math class (ගණිත පන්තිය) button in sub menu in order to do the math quiz. 3. Proceed with the enumerating quiz as stage 1 and click next button in dialog box after completion. 4. Proceed with the comparison objects and numbers quiz as stage 2 and click next button in dialog box after completion. 5. Proceed with the mathematical operation-based quiz covering the addition and subtraction quiz as stage 3 and click next button in dialog box after completion. 6. Display the results of each stages and average time to answer each question in result. 7. Click predict (හඳුනාගන්න) button. 8. Display whether children who suffer from dyscalculia or non- dyscalculia (සමත් or අසමත්).
Expected output	User Should be notify as dyscalculic child (අසමත්)
Actual output	User is notified as dyscalculia child (අසමත්)
Test result	Pass



❖ REFERENCES

[1] Dysgraphia Information Page. (n.d.). Retrieved July 21, 2020, from [Online]. Available:https://www.ninds.nih.gov/Disorders/All-Disorders/Dysgraphia-Information-Page [Accessed: 26-July- 2020]

[2] Equal rights to education for children with Learning Disabilities in Sri Lanka [Online]. Available: http://www.unicef.lk/everymind/ [Accessed: 26-July-2020]

[3] N. C. Zygouris *et al.*, "Screening for disorders of mathematics via a web application," *IEEE Glob. Eng. Educ. Conf. EDUCON*, no. April, pp. 502–507, 2017, doi: 10.1109/EDUCON.2017.7942893.

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[5] Universiti Tenaga Nasional, Universiti Tenaga Nasional. College of Information Technology, IEEE Computer Society, IEEE Malaysia Section, and Institute of Electrical and Electronics Engineers, "Conference Proceedings - 6th International Conference on Information Technology and Multimedia at UNITEN: Cultivating Creativity and Enabling Technology Through the Internet of Things, ICIMU 2014," *Conference Proceedings - 6th International Conference on Information Technology and Multimedia at UNITEN: Cultivating Creativity and Enabling Technology Through the Internet of Things, ICIMU 2014*. 2015.

[6] N. Beacham and C. Trott, "Screening for Dyscalculia within HE," *MSOR Connect.*, vol. 5, no. 1, Feb. 2005, doi: 10.11120/msor.2005.05010004.

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BSc (Hons) Degree in Information Technology (specialization in Software Engineering)

THE PROGRESS

Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition

INTRODUCTION

BACKGROUND

- Dyscalculia is a biologically based learning disability related to mathematic, numbers, and arithmetic skills.
- Centered around Neurological system.
- It is generally seen as mathematical equivalent of Dyslexia.
- According to British Dyslexia Association, Dyscalculia occur in 3 - 6% of population [1].
- There are 5 main types of Dyscalculia [2].
- Two types of them are focusing on here,

1. **Verbal Dyscalculia**
2. **Lexical Dyscalculia**



RESEARCH GAP

Features	1	2	3	4	Nena Shilpa
Number identification	✓	✓	✓	✓	✓
Voice recognition techniques	✗	✗	✗	✗	✓
Support Sinhala Language	✗	✗	✗	✗	✓
Improved UI/UX	✓	✓	✗	✓	✓
Keep track of progress	✗	✗	✗	✗	✓
Report Generation	✗	✓	✗	✗	✓
ML techniques	✗	✗	✗	✗	✓





1 - Mobile App for Dyscalculia Children in Malaysia [3]

2 - Mobile App Development on Detection and Diagnose of Learning Disability for Children [4]

3 - Intelligent System for Diagnosing Learning Disorders in Children [5]

4 - KidKanit

RESEARCH PROBLEM

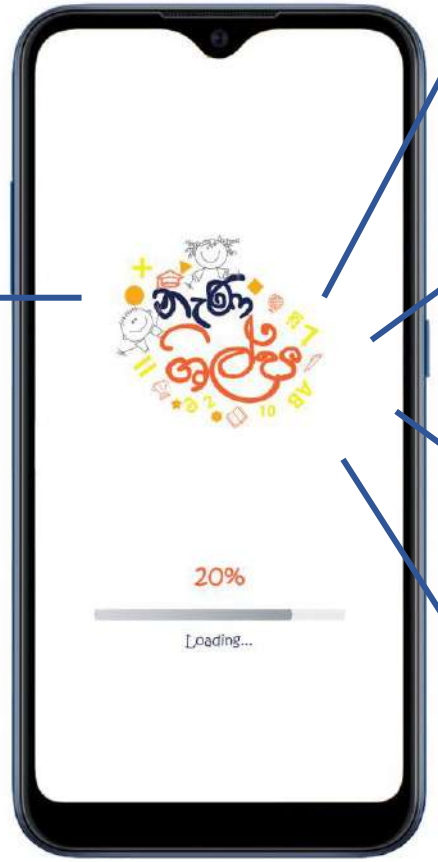


- How to Identify the risk for Lexical and Verbal Dyscalculia ?
- How to Provide activities to improve the condition ?
- How to Keep the track of progress ?

OBJECTIVES

Specific Objective

Identify the risk of Lexical and Verbal Dyscalculia and provide activities to improve the condition

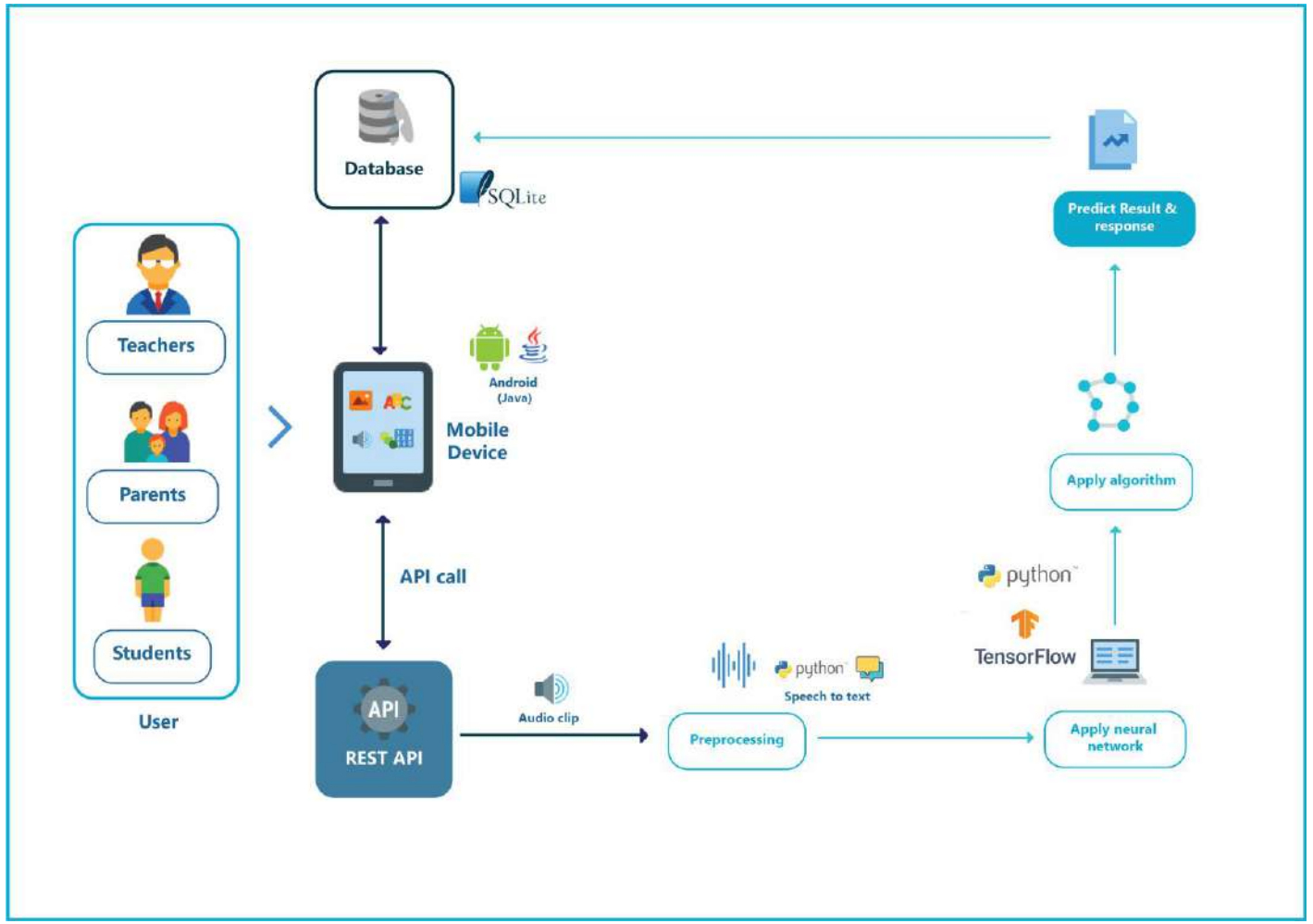


Sub Objectives

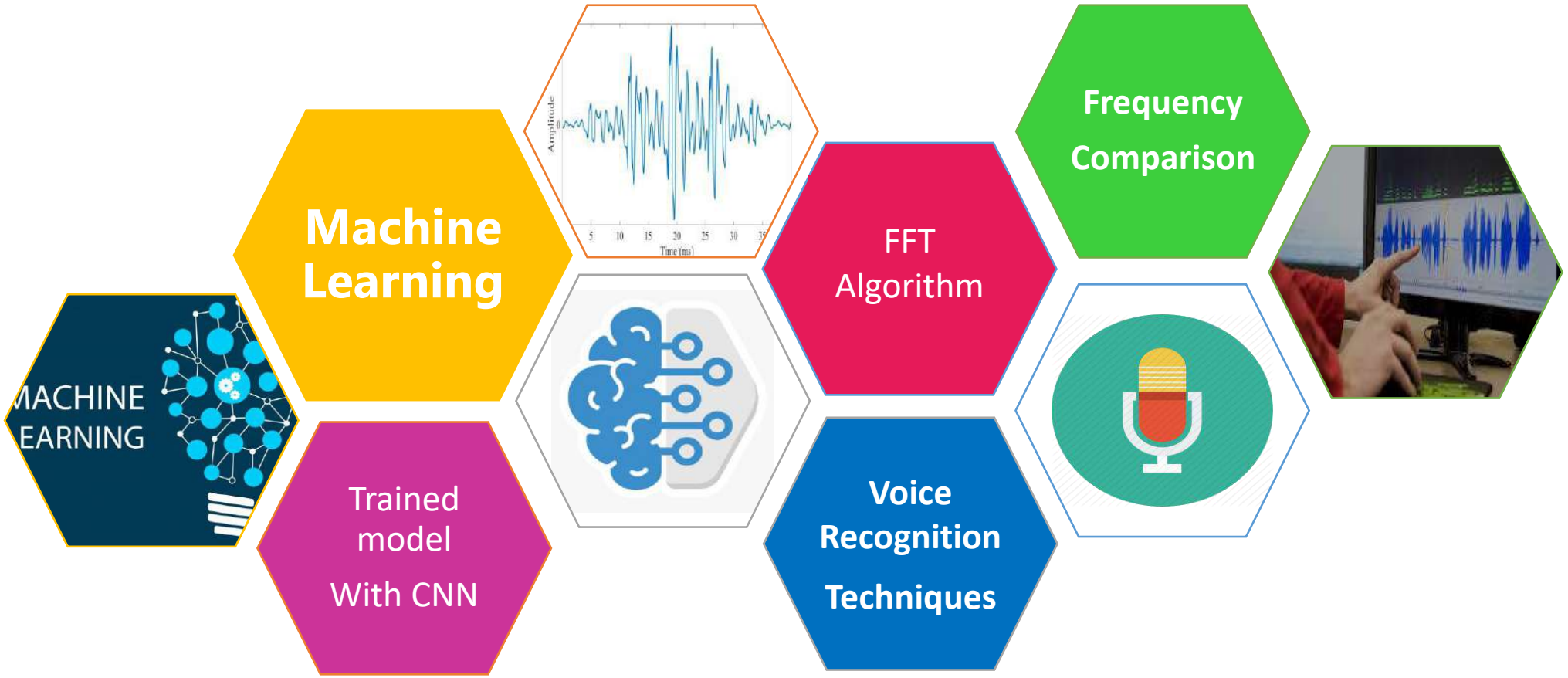
- Identify risk of Lexical Dyscalculia.
- Identify risk of Verbal Dyscalculia.
- Generate report with performed results
- Suggest remedies for identified user with risk.

METHODOLOGY

COMPONET OVERVIEW DIAGRAM



KEY PILLARS


Machine Learning
MACHINE
EARNING
Trained model
With CNN
Amplitude
Time (ms)
FFT Algorithm
Voice Recognition Techniques
Frequency Comparison

ACHIEVEMENTS



- Data gathering.
- UI/UX design.
- Tests for identify Verbal & Lexical Dyscalculia.
- Practice remedies.
- Save test records.
- 95 % CNN model accuracy.

ACHIEVEMENT EVIDENCES



Sub Menu



Test Select



Test Type 1



Test Type 2



Remedy Activity



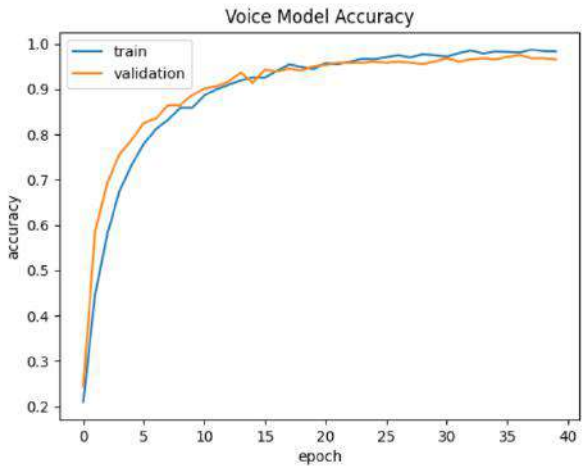
Previous Records

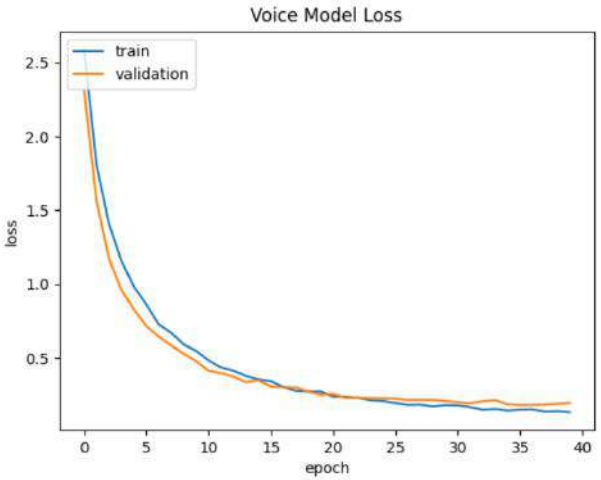
RESULTS & DISCUSSION

CNN Model Training Process

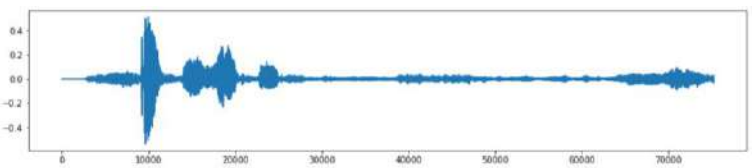
```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 128, 11, 64)       640
_________________________________________________________________
batch_normalization (BatchNo (None, 128, 11, 64)       256
_________________________________________________________________
max_pooling2d (MaxPooling2D) (None, 64, 6, 64)         0
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 62, 4, 32)         18464
_________________________________________________________________
batch_normalization_1 (Batch (None, 62, 4, 32)         128
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 31, 2, 32)         0
_________________________________________________________________
conv2d_2 (Conv2D)            (None, 30, 1, 32)         4128
_________________________________________________________________
batch_normalization_2 (Batch (None, 30, 1, 32)         128
_________________________________________________________________
max_pooling2d_2 (MaxPooling2 (None, 15, 1, 32)         0
_________________________________________________________________
flatten (Flatten)            (None, 480)               0
_________________________________________________________________
dense (Dense)                (None, 64)                30784
_________________________________________________________________
dropout (Dropout)            (None, 64)                0
_________________________________________________________________
dense_1 (Dense)              (None, 12)                780
=================================================================
Total params: 55,308
Trainable params: 55,052
Non-trainable params: 256
_________________________________________________________________
```

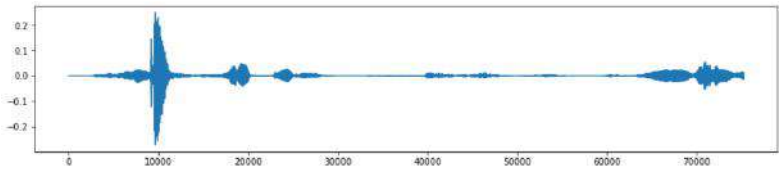
```
Test error: 0.19721291959285736, test accuracy: 0.9578713774681091
```

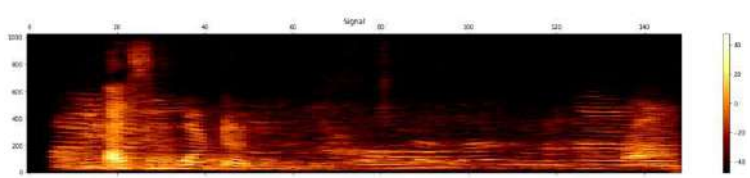




Audio Noise Reduction Process







Audio signal



Recovered Audio signal

TEST CASES

Test case ID	VT001
Test Case Scenario	Predict the risk of Verbal Dyscalculia – User pronounce number incorrectly.
Test input data	Audio input (.wav)
Test procedure	1. Select “ගණිත හුරුව” button on main menu. 2. Select “ගණිත පන්තිය” button from 1st sub menu. 3. Select “උත්තර කියවන්න” button from 2nd sub menu. 4. Listen voice guidance on how to perform the test. 5. Press Mic button and pronounce the displayed latter. 6. Press Next button to perform test again. 7. Perform test for five times. 8. Press Results button to view previous records.
Expected output	User should identify as a Verbal Dyscalculic user. (Risk above 50% accordingly)
Actual output	Identified user as a Verbal Dyscalculic user. (Risk above 50% accordingly)
Test result	Pass

TEST CASES

Test case ID	VT002
Test Case Scenario	Predict the risk of Verbal Dyscalculia – User pronounce number correctly.
Test input data	Audio input (.wav)
Test procedure	1. Select “ගණිත හුරුව” button on main menu. 2. Select “ගණිත පන්තිය” button from 1st sub menu. 3. Select “උත්තර කියවන්න” button from 2nd sub menu. 4. Listen voice guidance on how to perform the test. 5. Press Mic button and pronounce the displayed latter. 6. Press Next button to perform test again. 7. Perform test for five times. 8. Press Results button to view previous records.
Expected output	User should identify as a Non-Verbal Dyscalculic user. (Risk less than 50% accordingly)
Actual output	User identified as a Non-Verbal Dyscalculic user. (Risk less than 50%)
Test result	Pass

REFERENCES

[1] "Number Dyslexia: Everything to know about Dyscalculia - Number Dyslexia", *Number Dyslexia*, 2020. [Online]. Available: https://numberdyslexia.com/dyscalculia-aka-number-dyslexia/

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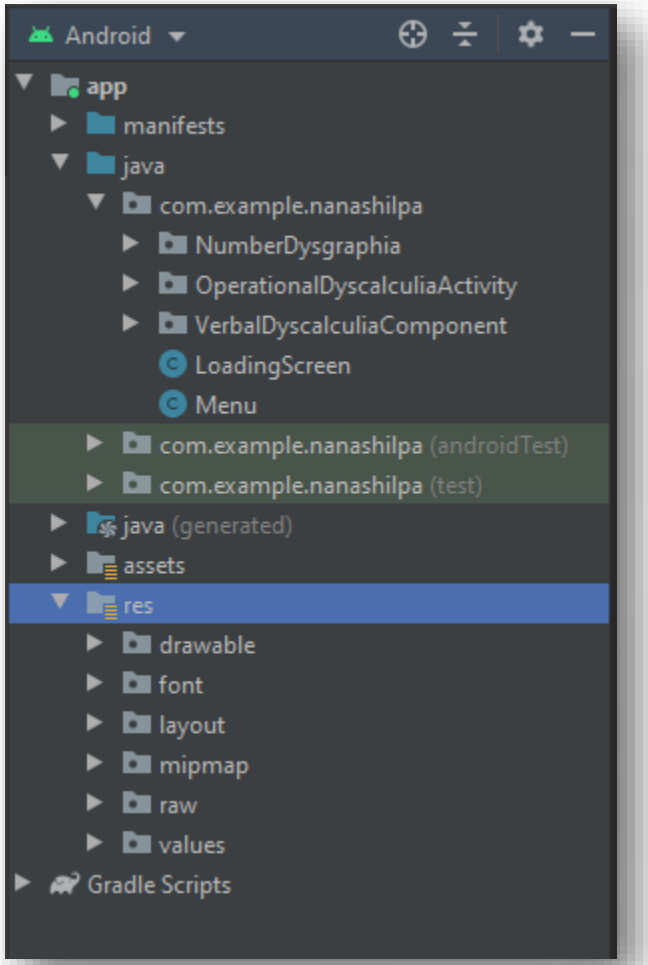


WHAT IS TO BE DONE ?



- ✓ Fine-tuning
- UI/UX design

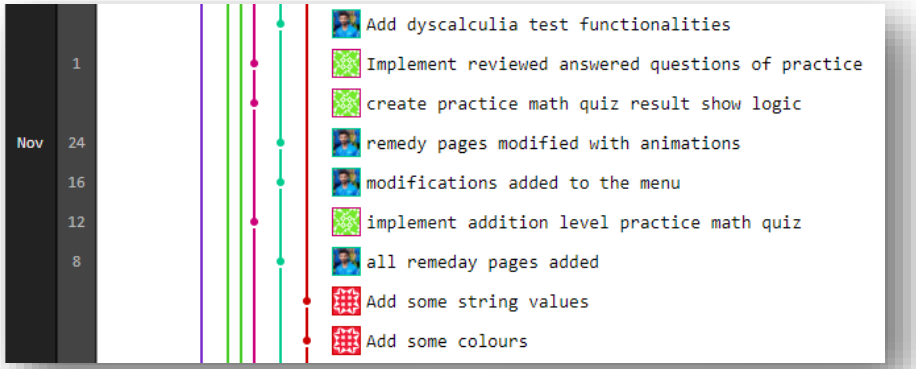
7. STANDARDS AND BEST PRACTICES



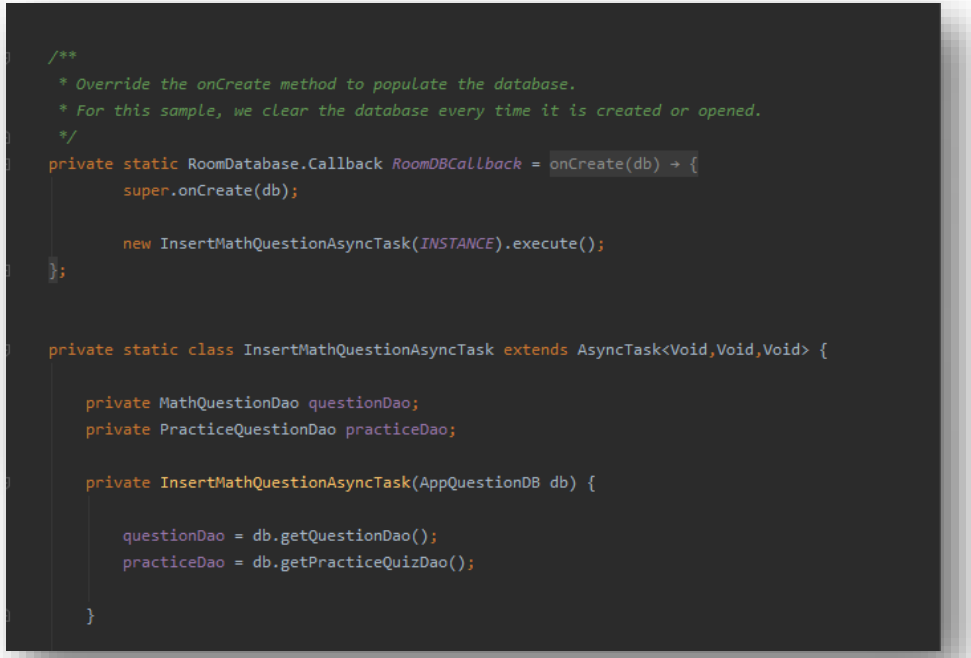
Packages for Component Organization



Industry standard for project architecture



Descriptive Source control



```
/**
 * Override the onCreate method to populate the database.
 * For this sample, we clear the database every time it is created or opened.
 */
private static RoomDatabase.Callback RoomDBCallback = onCreate(db) → {
        super.onCreate(db);

        new InsertMathQuestionAsyncTask(INSTANCE).execute();
};


private static class InsertMathQuestionAsyncTask extends AsyncTask<Void,Void,Void> {

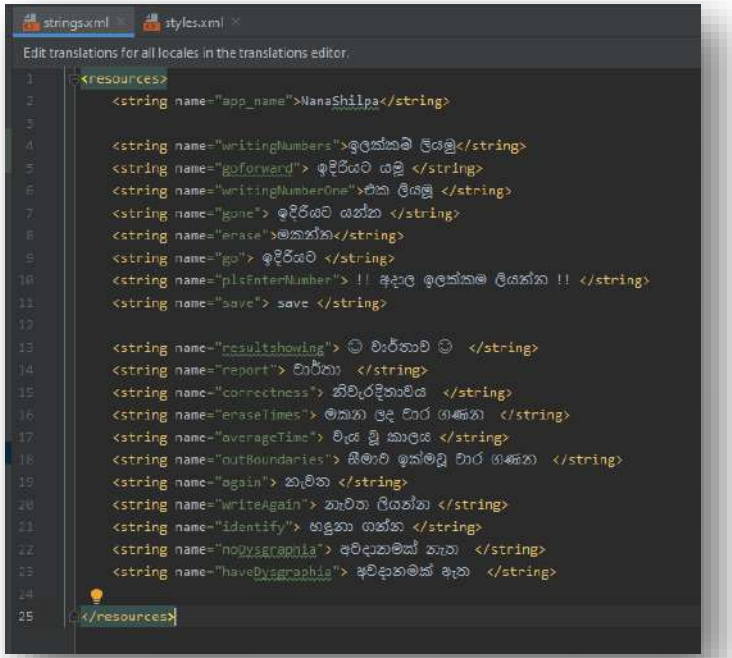
    private MathQuestionDao questionDao;
    private PracticeQuestionDao practiceDao;

    private InsertMathQuestionAsyncTask(AppQuestionDB db) {

        questionDao = db.getQuestionDao();
        practiceDao = db.getPracticeQuizDao();

    }
```

Increase code readability using developer comment



Resource file for better maintainability

8. NON-FUNCTIONAL REQUIREMENTS

Reliability

- Application server hosted in the cloud platform.

Performance

- Reduce no of API calls by processing data inside the Android application.



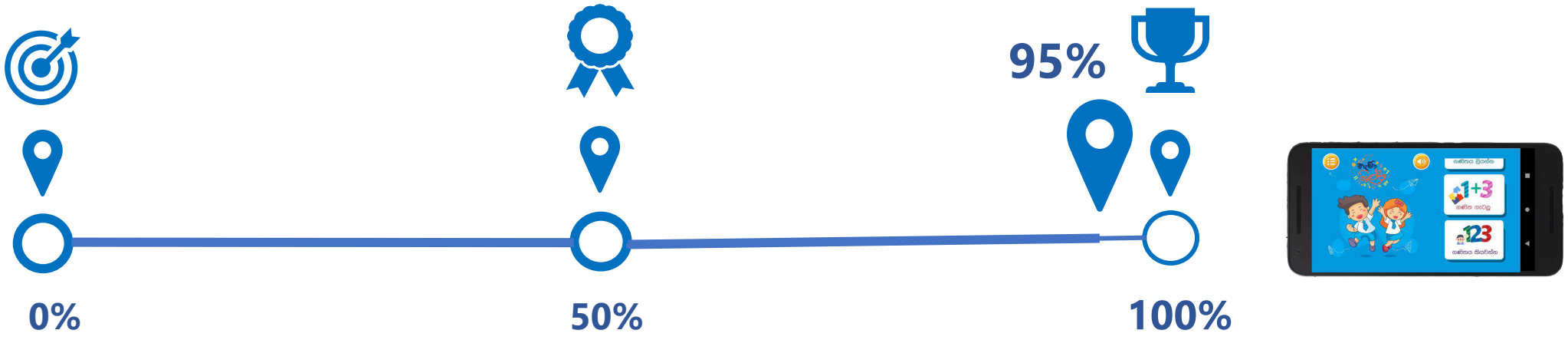
Usability

- Background voice guidance
- Optimal color combinations
- Process demo video guidance

Availability

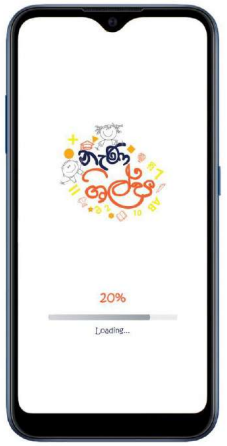
- Application server hosted in the cloud platform.

ADDRESSING THE FUNCTIONAL REQUIREMENT IN IMPLEMENTATION



- For the currently completed component implementation, all the functional requirements have been addressed which we had previously mentioned on the proposal presentation.

9. COMMERCIALIZATION ASPECTS



Using the proposed application, we hope to provide low-cost subscription plans considering each side.



For Schools
Rs. 1500

Permanent Subscription
Rs. 1000

Monthly Subscription
Rs. 300

Annual Subscription
Rs. 600

Low income Families.
Free Subscription.

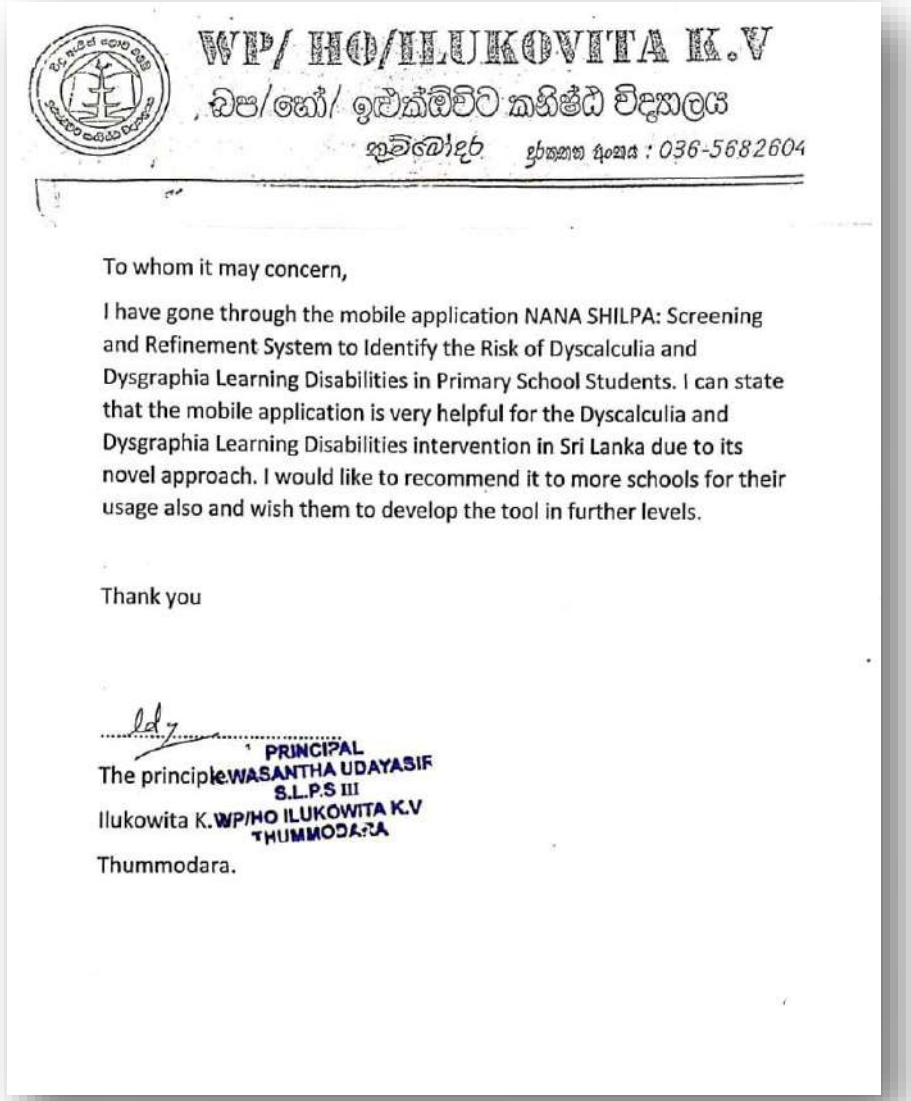
Child Orphanages.
Free Subscription

Social Media Commercialization.

10. BUDGET

Component	Amount (USD)	Amount (LKR)
Traveling fee for the possible consultation sessions and data gathering (Malabe, School selected for data extracting)	1.08	200.00
Internet charges (the development and technical information learning)	10.77	2000.00
Technical consultation charges (External technical information session)	5.4	1000.00
Total	17.25	3200.00

APPENDIX



WP/ HO/ILUKOVITA K.V
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තුම්මෝදර දුරකතන අංකය : 036-5682604

To whom it may concern,

I have gone through the mobile application NANA SHILPA: Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students. I can state that the mobile application is very helpful for the Dyscalculia and Dysgraphia Learning Disabilities intervention in Sri Lanka due to its novel approach. I would like to recommend it to more schools for their usage also and wish them to develop the tool in further levels.

Thank you

PRINCIPAL
W.A.S. UDAYASIRI
S.L.P.S III
WP/HO ILUKOWITA K.V
THUMMODARA

The principle
Ilukowita K.
Thummodara.



To whom it may concern,

I have personally gone through the mobile application NANA SHILPA: Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students which is implemented by the students Hewapathirana C.D.D. , Kalpani Sandupaba Abeysinghe A. D., Maheshani H.M.P. and Liyanage T.J.P.S.I can recommend this mobile application for children with special educational needs having to learn Dyscalculia and Dysgraphia.

Thank you

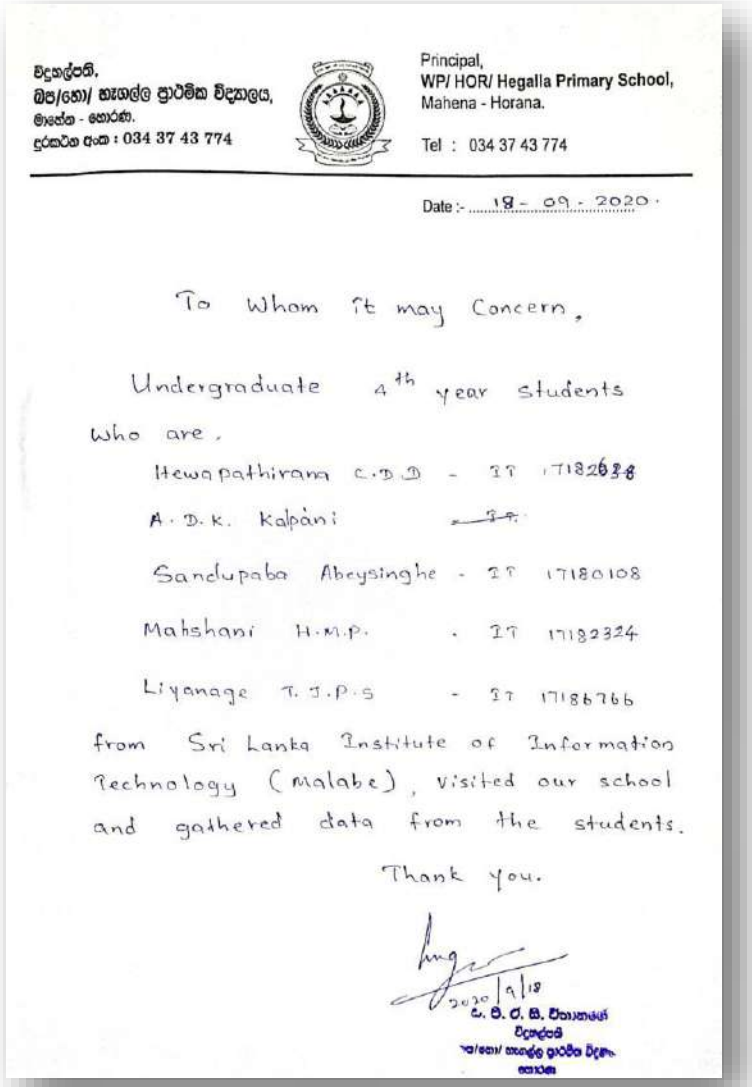
Dr. W. M. K. WANIGASINGHE
(MBBS)
S.L.M.C. Reg. No: 11614
MALABE

Dr. W.M. K. Wanigashinghe

Ministry of Health

Colombo 10

APPENDIX



විදුහල්පති,
බප/හෙ/ හැඟල්ල ප්‍රාථමික විද්‍යාලය,
මාහේන - හොරණ.
දුරකථන අංක : 034 37 43 774

Principal,
WP/ HOR/ Hegalla Primary School,
Mahena - Horana.
Tel : 034 37 43 774

Date :- 18 - 09 - 2020

To Whom it may Concern,

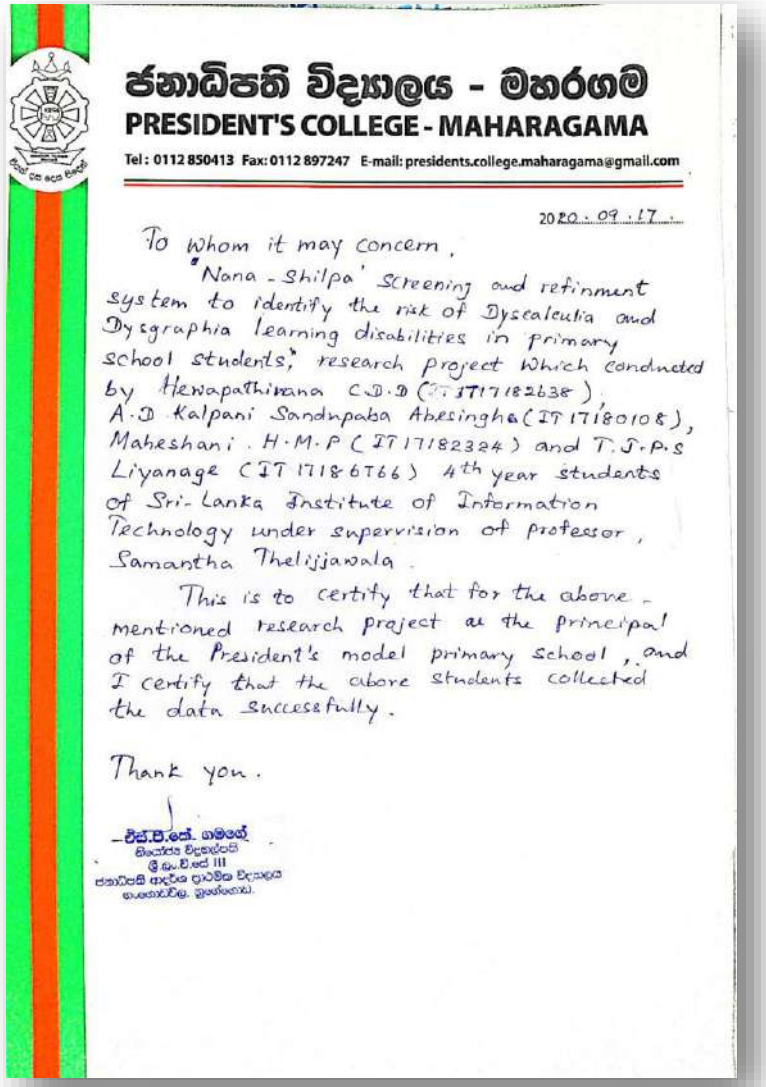
Undergraduate 4th year students who are,

Hewapathirana C.D.D - IT 17182638
A.D.K. Kalpani
Sandupaba Abeysinghe - IT 17180108
Mahshani H.M.P. - IT 17182324
Liyanage T.J.P.S - IT 17186766

from Sri Lanka Institute of Information Technology (Malabe), visited our school and gathered data from the students.

Thank you.

2020/9/18
ඒ. බී. ඩී. සී. විජයතුංග
විදුහල්පති
බප/හෙ/ හැඟල්ල ප්‍රාථමික විද්‍යා.
හොරණ



ජනාධිපති විද්‍යාලය - මහරගම
PRESIDENT'S COLLEGE - MAHARAGAMA
Tel : 0112 850413 Fax: 0112 897247 E-mail: presidents.college.maharagama@gmail.com

2020 . 09 . 17

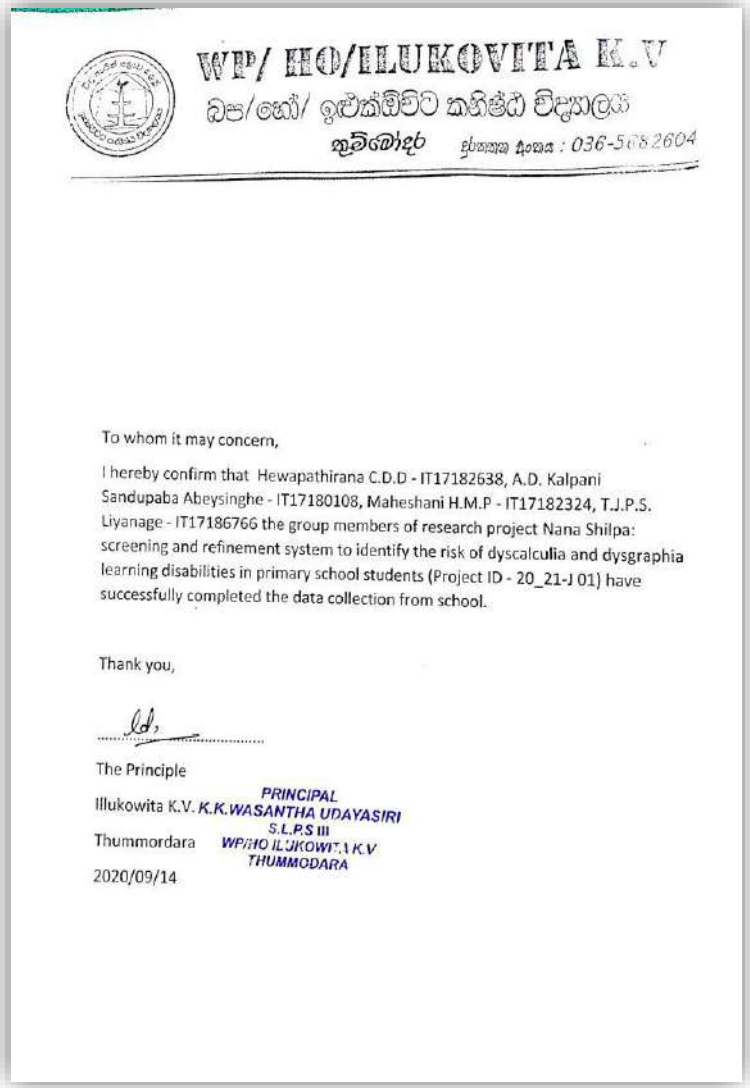
To whom it may concern,

'Nana - Shilpa' screening and refinment system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students, research project which conducted by Hewapathirana C.D.D (IT 17182638), A.D Kalpani Sandupaba Abesingha (IT 17180108), Maheshani . H.M.P (IT 17182324) and T.J.P.S Liyanage (IT 17186766) 4th year students of Sri-Lanka Institute of Information Technology under supervision of professor, Samantha Thelijjawala.

This is to certify that for the above - mentioned research project at the principal of the President's model primary school, and I certify that the above students collected the data successfully.

Thank you.

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විදුහල්පති
ජී.ඉ.වි.සේ III
ජනාධිපති ආදර්ශ ප්‍රාථමික විද්‍යාලය
මහරගම. මුලණගොඩ.



WP/ HO/ILUKOVITA K.V
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තුම්මොදර දුරකථන අංකය : 036-5682604

To whom it may concern,

I hereby confirm that Hewapathirana C.D.D - IT17182638, A.D. Kalpani Sandupaba Abeysinghe - IT17180108, Maheshani H.M.P - IT17182324, T.J.P.S. Liyanage - IT17186766 the group members of research project Nana Shilpa: screening and refinement system to identify the risk of dyscalculia and dysgraphia learning disabilities in primary school students (Project ID - 20_21-J 01) have successfully completed the data collection from school.

Thank you,

The Principle
Illukowita K.V.
Thummordara
2020/09/14

PRINCIPAL
K.K.WASANTHA UDAYASIRI
S.L.P.S III
WP/HO ILUKOWITA K.V
THUMMODARA

APPENDIX





DEMOSTRATION







Q & A.



THANK YOU